picBunch v0.7.0: a free-space FFT convolution for the initial field of a uniformly moving particle bunch, and its measured cost and accuracy

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Release v0.7.0

1 Introduction

When a particle-in-cell (PIC) simulation starts, the bunch of particles must carry an initial electromagnetic field, or the first push already acts on the wrong force. The need is sharpest for laser wakefield acceleration simulations, where a driven bunch moves relativistically through the wake [Esarey et al., 2009] and the first push acts on whatever field the start state supplies. The field of a moving bunch in free space is known in closed form for each point charge, but a simulation reads that field on a staggered mesh with thousands of observation points, so it is computed by a fast sum over the source charges rather than by evaluating the closed form everywhere. We compute it as a free-space FFT convolution of a TSC deposit with the sampled TSC-convolved point-charge kernel [Birdsall and Langdon, 1985, Hockney and Eastwood, 1981]. The field is the present-time field of uniformly moving charges, with no radiation term and no history [Jackson, 2007, Landau and Lifshitz, 1975]. This paper gives the exact discrete recipe a re-implementer needs, and it measures the cost and the accuracy of `picBunch v0.7.0` on a V100, an H100 and a gfx1100, naming the reference for every error statement. The release is driven by a single TOML configuration (input file, destination, config; Section 3.1), and it carries the capabilities the measurements below exercise: an opt-in near shell with a fast closed-form evaluator, MPI component-parallel mesh and tile-parallel near shell with bitwise equality across rank counts, a tolerance-derived far-tail margin for zero fill, and a float32 openPMD checkpoint that restarts under a stock PIConGPU [Burau et al., 2010, Huebl et al., 2015].

The established workaround avoids computing the initial field altogether: one starts from a field-free state and marches the bunch forward, letting the fields build up in the first few steps. The present-time field of a uniformly moving bunch has no transient, so this history has no physical basis. We measure how well the discrete method reproduces that field and how much it costs, and we state the conditions under which the measured error is largest (Section 5).

2 Field definition

The computed quantity is a single linear integral. For a bunch of source charges at positions $\mathbf{x}_p$ with signed charge q_p, and a shared dimensionless velocity $\mathbf{u} = \mathbf{p}/(mc)$, the field at a staggered observation point $\mathbf{x}_{\text{obs}}$ is the TSC-convolved uniformly-moving point-charge field

$$F_c(\mathbf{x}_{\text{obs}}) = \sum_p q_p \int_{[-1.5,1.5]^3} W(\mathbf{s})\, K_{\text{point},c}\big(\mathbf{x}_{\text{obs}} - \mathbf{x}_p - \mathbf{s}\,\mathbf{cell},\ \mathbf{u}\big)\, d\mathbf{s}^3, \tag{1}$$

where $W(\mathbf{s}) = W_x(s_x)W_y(s_y)W_z(s_z)$ is the separable TSC shape (Section 3) and K_{point} is the per-unit-charge field of one uniformly moving charge. Because the integral is linear in the source charge, the sum over particles is exact; the whole design is a fast evaluation of this one integral.

2.1 The moving point kernel

The kernel is the present-time Liénard–Wiechert field of a uniformly moving charge, in the present-position form of Jackson and of Landau and Lifshitz (this work). instantaneous position. In the lab

frame, with $\mathbf{R} = \mathbf{x}_{\mathrm{obs}} - \mathbf{x}_{\mathrm{src}}$,

$$\gamma = \sqrt{1 + |\mathbf{u}|^2}, \qquad \boldsymbol{\beta} = \mathbf{u}/\gamma, \qquad \mathrm{invg}^2 = 1/(1 + |\mathbf{u}|^2) = 1 - \beta^2, \tag{2}$$

$$D = |\mathbf{R}|^2 \,\mathrm{invg}^2 + (\mathbf{R}\cdot\boldsymbol{\beta})^2, \tag{3}$$

$$\mathbf{E}_n = \mathrm{invg}^2 \,\mathbf{R}/D^{3/2}, \qquad \mathbf{B}_n = \boldsymbol{\beta} \times \mathbf{E}_n. \tag{4}$$

For $\mathbf{u} = \mathbf{0}$ exactly, $\mathbf{E}_n = \mathbf{R}/|\mathbf{R}|^3$ and $\mathbf{B}_n = \mathbf{0}$; no ε is added to any denominator. The kernel is homogeneous of degree -2 in $\mathbf{R}$. A bunch with a velocity spread needs one convolution per velocity class (Section 6.16); the validated scope is a single shared $\mathbf{u}$.

2.2 The unit chain

The reference normalizes the per-unit-charge field as $E_n = 4\pi\varepsilon_0\, E/q$ and $B_n = 4\pi\varepsilon_0 c\, B/q$, so $B_n = \boldsymbol{\beta}\times E_n$ (this work). The mesh produces the per-unit-charge field with $\mathbf{R}$ in metres, already summed over the deposited charge q_p in coulombs. The chain to physical SI is

$$E_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0), \quad B_{\mathrm{SI}}[c] = F_c[c]\, h^3/(4\pi\varepsilon_0 c), \tag{5}$$

with $\varepsilon_0 = 8.8541878128 \times 10^{-12}$ F/m and $c = 299792458$ m/s. The h^3 factor (Section 3) is the Riemann volume of the sampled continuum integral

2.3 The TSC shape

W is the quadratic B-spline, with argument in cell units, support $[-1.5, 1.5]$ per axis, and unit integral per axis, the standard TSC shape of the PIC literature [Birdsall and Langdon, 1985, Hockney and Eastwood, 1981]:

$$W(x) = \begin{cases} 3/4 - x^2 & |x| < 1/2, \\ \frac{1}{2}(3/2 - |x|)^2 & 1/2 \le |x| < 3/2, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

3 Method

3.1 The operator interface

`v0.7.0` is driven by `picBunch -input bunch.h5 -destination <checkpoint> -config params.toml`, with every unit and conversion parameter in the TOML file. The `[picongpu]` section carries `unit_length`, `unit_e`, `unit_b`, `dt`, `step`, `precision`, `species`; `[domain]` carries `cells`, `cell_size`, `origin`; `[placement]` carries `offset_unit` and the `left`/`front`/`top` fractions; `[accuracy]` carries `tolerance` and the near-shell settings, `near_shell none`/`exact`/`fast`/`fast-gpu` plus `near_shell_radius` and `near_shell_tile`. A missing conversion key causes a clear error at start-up. The input bunch is centred by its charge-weighted centroid and placed at the requested offset from the domain origin; a placement leaving the bunch outside the domain is rejected at start-up. `patch_count` in `[placement]` splits the species into one particle patch per output rank, with cumulative, disjoint `numParticlesOffset` ranges, for multi-rank restarts (Section 7).

The default `precision` is uniform `float32` for every field and particle record, and such a checkpoint restarts under a stock single-precision PIConGPU; `float64` is a uniform opt-in that records a warning attribute because stock PIConGPU cannot load double records. The `charge` and `mass` records are written rather than omitted, and the species is a single particle patch, `patch` = `"bbox"` or `"domain"`.

The mesh evaluates Eq. (1) by a discrete deposit, a sampled kernel, and a free-space FFT convolution. Let N be the bunch-box nodes per axis, $M = \mathrm{pad}\cdot N$ the padded FFT grid per axis, h the node spacing in cell units (node i sits at $ih\,\mathbf{cell}$), and $\mathbf{cell} = (dx, dy, dz)$. All precision-critical work is in double

3.2 Deposit

Each source charge is deposited onto the zero-padded M^3 grid with the TSC shape, at the cell-unit spline argument $W((\mathbf{x}_i - \mathbf{x}_p)/\mathbf{cell})$, which is correct for any h (this work). The result $\rho[i] = \sum_p q_p\, W(ih - \mathbf{p}_{\mathrm{cell}})$ lives in the N^3 sub-box of the padded grid, indexed with M-stride; the rest is zero. A fixed reduction order is required for MPI-layout invariance; the single-rank path uses `atomicAdd`.

3.3 Kernel sampling

The six TSC-convolved kernels are sampled on the same padded M^3 grid, one thread per FFT point, with the centered (negative-wrapped) index $d_a = a$ for $a < M/2$ else $a - M$. For each of the six components c, the kernel is evaluated at the physical separation

$$\mathbf{R} = (d_a\, h + \delta_{c,a})\, \mathbf{cell}, \tag{7}$$

where δ_c is the Yee staggered offset (Section 3.5). The kernel is therefore sampled at the same node spacing h as the deposit, with the shape one grid cell of its own axis wide (this work). Non-finite samples are set to zero to avoid $0 \times \infty$ in the FFT.

3.4 FFT convolution and normalization

The convolution is a free-space, zero-padded FFT convolution, with the discrete convolution theorem applied in the unnormalized form of the vendor library documentation [Press et al., 1986, NVIDIA Corporation, 2024]. The vendor plan is built once, unnormalized, and reused for all thirteen transforms: one forward for ρ, six forwards for the kernels, and six inverse transformations. The unnormalized round trip is scaled by M^3; the code divides by $h^3 M^3$ in the read-out

Padding. The grid must be zero-padded to pad ≥ 2 so that the cyclic convolution does not wrap [Press et al., 1986]. With pad $= 1$ the cyclic convolution wraps and a 9.275×10^{-3} periodic-image artefact appears for a single stationary particle in a 20^3 box (this work). For a single stationary particle the pad $= 2$ result agrees with the direct non-cyclic sum to 3.811×10^{-16} and the pad $= 3$ result to 1.905×10^{-16}. At pad ≥ 2 the error is at machine precision ($\leq 10^{-15}$). The artefact grows for beamed sources: at pad $= 1$ the 100-source $\gamma = 10$ cloud shows a 7.5×10^{-1} worst-component error

DC and Nyquist. For the R2C/C2R half-spectra the DC bin and the seven self-conjugate even Nyquist corners must be exactly real; the intermediate Nyquist planes are Hermitian-paired, not real (this work). The R2C/C2R path matches the C2C path to 1.665×10^{-16} over all six components.

The h^3 factor. The sampled convolution evaluates the continuum integral of Eq. (1) by a Riemann sum whose nodes are spaced h cells apart, so each sample covers a physical volume h^3 (in cell units); the read-out therefore applies the factor $h_x h_y h_z / (M_x M_y M_z)$

3.5 Read-out

The six field components live at the staggered Yee positions, in cell units (this work):

$$\begin{array}{llllll} E_x & (0.5, 0, 0) & E_y & (0, 0.5, 0) & E_z & (0, 0, 0.5) \\ B_x & (0, 0.5, 0.5) & B_y & (0.5, 0, 0.5) & B_z & (0.5, 0.5, 0). \end{array} \tag{8}$$

The kernel is sampled at these offsets, so the convolution is already evaluated on the Yee grid. The read-out compacts the M^3 FFT result to the N^3 bunch box and applies the h^3/M^3 scale and the SI factor (Section 2). No further interpolation is needed.

3.6 Scale invariance

The kernel is homogeneous of degree -2, the deposit shape width equals the node spacing, and the Yee offsets are in cell units, so every relative metric depends only on r/h, the distance in cell units (this work). At a fixed distance in cells the error is therefore independent of h: refining h with the distance held in cells changes nothing, and the measured relative errors at $h = 1, 1/2, 1/4, 1/8$ are identical to machine precision for $\gamma \in \{1, 10, 100\}$, aspect 1 and 4. At a fixed physical distance r, refining h increases r/h and does reduce the error, at the cost of M^3 FFT work in $1/h$. The controlling length for a beamed source is the beaming cone width d/γ: where that width is sub-cell, the error is set by the cone, not by the mesh, and refinement stops paying off (Section 6.13).

3.7 The near shell and the fast evaluator

The default mesh is the whole field. `v0.7.0` carries an opt-in near shell (`[accuracy]` in the TOML config, Section 3.1), with a radius and a tile A distributed-FFT option (Section 6.10) makes the output field slab-local. It computes

$$F = F_{\text{mesh}}(\text{all}) - F_{\text{mesh}}(\text{near}) + F_{\text{near}}(\text{near}), \tag{9}$$

where the subtraction uses a per-tile local zero-padded FFT of the near sources (padding 2) and the near term is evaluated per (source, node, component) by one of two backends selected by `near_shell`: `exact`, a vendored adaptive FP64 TSC-convolution quadrature, or `fast`, the closed-form evaluator below. The default remains `none`; the radius is in mesh nodes of the compute grid, and the envelope it must cover depends on h (Section 6.5). The fast backend is the accurate one: its measured accuracy and cost are in Sections 6 and 6.8 (this work). The `exact` evaluator shares its vendored adaptive Gauss–Legendre quadrature with the independent driver reference, and that path overestimates moving oblique in-support points by up to 7× at a specific node-particle configuration (the forward case) while agreeing with the arbiter to within a few percent for the slice and large-domain geometries, so `exact` is retained for comparison rather than as a moving in-support truth (Section 6.5, this work).

The fast evaluator splits the in-support integral of Eq. (1) into a singular part and a smooth remainder. The singular part is the eight-term `asinh` box of the kernel's divergence, O(1) per pair, built from the double integral

$$\begin{aligned} T(A,B,u) &= \int_0^A \int_0^B \frac{\mathrm{d}X\,\mathrm{d}Y}{\sqrt{u^2+X^2+Y^2}} \\ &= \begin{cases} A\,\mathrm{asinh}(B/A) + B\,\mathrm{asinh}(A/B), & |u| < 10^{-300}, \\ A\,\mathrm{asinh}\big(B/\sqrt{u^2+A^2}\big) + B\,\mathrm{asinh}\big(A/\sqrt{u^2+B^2}\big) & \\ \quad -u\,\mathrm{atan}\big(AB/(u\sqrt{A^2+u^2+B^2})\big), & \text{otherwise.} \end{cases} \end{aligned} \tag{10}$$

with $A, B > 0$ (each box face has 8 corners, 24 such calls per component, so the eight-term box is O(1) per pair). The form is even in u, so the $|u| \to 0$ branch must fire on $|u|$, not u (a negative u is a field point beyond a box face) (this work). Verified against a converged midpoint trapezoid reference (-2.6348 converged; the naive trapezoid -2.6333 is 1.5×10^{-3} low, a trapezoid singularity artefact (this work)). The primitive is implemented in the fast evaluator header `include/picBunch/detail/FastNear.hpp` and the device reference `tools/closed-form/gpu_extract.cu`. The remainder is a plain tensor Gauss–Legendre quadrature on the subtracted integrand $(S - W^*)K$ over the TSC knot panels, each axis split at the TSC knots and at the anchor point s^*; the default order is 5, 8000 nodes per in-support pair (this work). For a moving bunch the box is γ-stretched for the axial direction, and the oblique direction uses a rest-frame octant quadrature of the singular box at order 8, 4096 nodes, plus the same remainder. Host and device use the same algebra.

The assembly is exact. Single-source tests at a tile centre and a tile boundary give an assembly residual of 2×10^{-12} to 1.5×10^{-11} SI (4×10^{-16} relative), flat with distance. A local FFT of the near set against the global FFT agrees at 10^{-15} for margins 0, 2, 4 and a 100-particle cloud, and boundary accounting shows the same near set feeding the exact sum and the subtracted mesh (field identity 3.6×10^{-13}). A direct reproduction with 100 particles at radius 3 gives an assembled field exact to 4.6×10^{-13} over the whole grid, while the plain mesh on the same plane is wrong by 5.6×10^{-1}

The fast backend changes no far number: the mesh-only field from the fast and the exact backends is bitwise identical (same md5), and the total field outside the near tiles is bitwise equal, so the tile subtraction stays intact The seam jump excess at the fast/exact crossing, on a 16^3 grid, $R = 3$ cells, tile 4, cells 10^{-7} m, is 2.03×10^{-10} per component for a stationary single particle (2.02×10^{-10} on $E_x/E_y/E_z$, zero on B) and 2.8×10^{-10} for the 64-source cloud.

3.8 Bunch and grid configurations

Every case measured in the paper is listed in Table 1 with the bunch and the grid, and the section that uses it. The entries are the recorded configuration values of this work: the mid-bunch slice cases (bunches generated on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, centroid $(32, 32, 16)$, $\sigma = 6$ cells per axis); the large-domain cases; the cost and accuracy cases; the γ landscape and the aspect-4 rows;

the multi-velocity cases; and the divergence case (whose independent reference is pending, so no row is included). The generation line for the mid-bunch bunches is `generate_bunch.py -model gamma -gamma G -seed 42 -grid 64 64 32 -cell-size 1e-7 1e-7 1e-7 -centroid 32 32 16 -sigma 6 6 6 -weights uniform` (this work). Entries with no recorded value are written as "not recorded"; nothing here is inferred. The total charge is N times $-e$ for the electron bunches with one unit weight, where $e = 1.602176634 \times 10^{-19}$ C; the kinetic energy is $(\gamma - 1)\, m_e c^2$ with $m_e c^2 = 510.999$ keV.

The mid-bunch bunches are mono-energetic: the read-back records $\gamma_{\min} = \gamma_{\max} = 1$ for the $\gamma = 1$ bunch and $\gamma_{\min} = \gamma_{\max} = 10$ for the $\gamma = 10$ bunch. The earlier multi-velocity cases (48^3, 512 sources) were superseded by the v0.9.1 re-measurement on a 64^3 grid with 200 sources (fluctuation bunches, 10^{-7} m cell, `hal` A30); the v0.9.1 numbers in Section 6.16 replace the earlier binning-error table. The earlier campaign bunch files were unavailable; the per-particle sample was regenerated from the surviving driver block (seeds 100, 103) on `rosi_v100`; the LCG state and draw count are identical to the campaign, and only last-bit `libm` differences between hosts are possible, so the sampled distributions are statistically identical.

3.9 MPI parallelism

`v0.7.0` writes a global output domain larger than the compute box (this work), and computes the field in parallel across ranks The written extent is set by `-global-grid` and `-global-origin`. The global array is never allocated: each rank owns a contiguous z-slab split into tiles of about 128 MiB, zero-filled, then filled with the compute-box overlap, the rest exact 0.0. All ranks open one openPMD series [Huebl et al., 2015] with `MPI_COMM_WORLD` and write disjoint `storeChunk` regions with collective flushes, so the output is a single dataset, not per-rank files. A `-deterministic-deposit` flag gives a serial host deposit for bitwise comparisons.

The compute is parallel in two ways, neither of which is a domain-decomposed FFT (this work). The mesh runs six independent component convolutions: every rank deposits and transforms the same ρ spectrum (replicated), rank r evaluates the components c with $c \bmod \text{size} = r$, and one `MPI_Allreduce` sums the six arrays. The near shell partitions the output tiles by linear tile index; each rank builds its own near-source list and local subtraction FFT for its tiles and writes only those corrections, and one `MPI_Allreduce` sums the correction arrays. The bunch is read once on rank 0 and broadcast, not redistributed. Because the mesh components are disjoint and each near correction element is owned by exactly one rank, the all-reduce order cannot change a value: the outputs are bitwise identical at 1, 2 and 4 ranks (Section 4.1).

3.10 The far-tail margin

For a global domain that extends beyond the compute box, the zero fill must not start until the field is below tolerance. `v0.7.0` places that boundary at the smallest centre distance d where

$$E_{\text{tail}}(d) = \gamma_{\max}\, k \left(\frac{|Q|}{d^2} + \frac{2|\mathbf{p}|}{d^3} + \frac{3.5\, \|D\|_2}{d^4} \right) \le \text{tolerance} \cdot E_{\text{peak}}, \tag{11}$$

with Q the total charge, $\mathbf{p}$ the dipole, $D_{ij} = \sum q\,(3x_i x_j - |\mathbf{x}|^2 \delta_{ij})$ the traceless quadrupole from the actual particle positions, $\|D\|_2$ its largest absolute eigenvalue, and $\gamma_{\max} = \max \sqrt{1 + |\mathbf{u}|^2}$ (this work). Beyond a measured switch radius the writer instead evaluates the multipole field directly (`FarTail.hpp`), so the shipped file has no interior zero fill. The margin is reported at run time as a bound in V/m.

For the documented 20-cell, 4096-electron bunch (source radius $R = 16.85$ cells, peak 45722.5 V/m, tolerance level 45.72 V/m), the quadrupole margin is 350 cells, which exceeds the 2^{28} padded-node cap, so the code falls back to the measured far-tail switch. Table 4 gives the switch error: the worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum in 13 directions

A shipped 512^3 output sampled outside the compute box confirms the switch in practice: 36 nodes including corners, worst $|\mathbf{E}_{\text{file}} - \mathbf{E}_{\text{direct}}| = 2.8 \times 10^{-3}$ V/m against the 45.7 V/m tolerance level (6.1×10^{-5} of it), and every sample nonzero, i.e. no interior zero fill

4 Measured scaling

All timings are on one host, `rosi_v100` (node `gv002`), exclusive Slurm allocation, pinned to GPU 0. Build: CUDA 12.6, `PICBUNCH_ARCH=70`, Release, padding 2, double. Each configuration ran three timed

repeats after one untimed warm-up plus one device-level warm-up; stage values are medians, the total is min/median/max. The four $\gamma = 10$, $N = 2 \times 10^6$ repeats give total medians 60.35 to 64.76 ms, about 7% scatter; the 64.81 ms row is the $\gamma = 100$ run. These compute-only totals exclude the checkpoint writer; the end-to-end totals with the writer are in Table 6, a different measurement path.

Reproducibility of Table 5. The matrix is the `picBunchBench` sweep, run with picBunchBench--sweep--repeat3--csvmatrix.csv (this work). The input bunches are not checked into the repository; the sweep configuration is the generator. The fast GPU kernel tools/closed-form/gpu_extract.cu is part of the shipped source The H100 runs used CUDA 13.4.

Correctness anchor. Every timing run is anchored by the Coulomb limit ($\mathbf{u} = 0$): one particle at a node, far nodes at $d = 8, 16, 24, 32$ along all six axis directions. The mesh field against the direct TSC stencil sum of the analytic Coulomb kernel has worst relative error 4.88×10^{-14} (max absolute 2.28×10^{-25}), passing the 10^{-12} threshold This confirms the zero-padded FFT convolution reproduces the TSC-convolved Coulomb field to roundoff.

Fig. 1 sweeps one axis at a time: the top panel fixes the 20-cell box and varies N, the bottom panel fixes $N = 2 \cdot 10^6$ and varies the box. The scaling is in two regimes, with the crossover set by the box, not the particle count. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows while the FFT, kernel, and read-out stay under 10 ms at the 20-cell box; the total is flat across $\gamma = 1, 10, 100$ (64.9/64.4/64.8 ms). At large boxes the kernel and FFT scale with M^3: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× because the deposit stays constant. The per-h totals are 64.76, 69.27, and 151.51 ms for $h = 1$, 1/2, 1/4 (padded array $M = 60/100/180$, 2×10^6 particles, 20-cell box). On the `H100` (`jureca-h100`, `sm_90`), the same 2×10^6, 20-cell, $h = 1$ mesh is 0.052 s compute (deposit 34.4 ms, kernel 1.08 ms, FFT 7.44 ms, read-out 9.02 ms; medians of three), and 0.755 s CLI wall including the particle read and the HDF5 write.

No configuration came near the 32 GB device limit; the largest exact device allocation was 468.7 MiB. Memory scales with the padded bunch box, not the global field.

Table 6 is the same pipeline with the writer in the loop. The deposit is still linear in N (2.70 → 31.56 ms from 10^4 to 10^6), and the FFT is still box-bound and flat. The writer is the dominant added term: 11.29 ms at 10^4 and 7.54 ms at 10^6. The $N = 10^6$ total is 47.15 ms with the writer, versus 61.92 ms in the compute-only Table 5; the two campaigns differ in measurement path and timing boundary, so neither is the other's total plus the writer

4.1 MPI rank scaling

The rank-parallel compute (Section 3.9) and the chunked writer were measured on an idle `rosi_v100` window, medians of three runs (this work). The node was checked idle before the block (no CPU worker, 1-min load below 2), but each rank block starts from the preceding measurement, so the 1-min load rises within the block: the near-shell block starts at 1.9, 8.4 and 11.8 for 1, 2 and 4 ranks (mesh block 1.4–1.5, write block 1.1–1.6); no foreign process was active during the window (this work). Correctness was verified at 1, 2 and 4 ranks on a 128-particle, 32^3, cell 8×10^{-7} m, tolerance 10^{-3}, near-shell $R = 1.5$, tile 8 problem: the mesh and the near shell are bitwise identical at 1-vs-2 and 1-vs-4, all six components, $\max_{\text{abs diff}} = 0.0$.

Fig. 2 shows the speedup for the three parallel pieces. The mesh is component-parallel: the replicated deposit (307–332 ms, flat in rank count) and the replicated source-spectrum FFT cap it, so the speedup plateaus at 1.20× at both 2 and 4 ranks. The near shell is tile-parallel: the static round-robin tile split plus the all-reduce of the full correction array leave static-tile imbalance that plateaus it at 1.34× at 4 ranks. The writer is a collective HDF5 write: it improves to 1.71× at 2 ranks and falls to 0.48× at 4 ranks, where the collective flush and metadata traffic contend on the shared filesystem, an I/O limit rather than a compute limit.

5 Measured error structure

The measured error depends on the observation point, the direction relative to the beam, and γ. We report the regimes in which the error is largest, together with the configuration and reference for each number.

The fast near backend, against the independent continuum, is 7.99×10^{-4} stationary interior, 1.96×10^{-5} face, and 1.57×10^{-7} outside; 1.69×10^{-4} moving axial interior, 1.11×10^{-4} face, 2.12×10^{-6} outside; and 8.15×10^{-2} moving oblique interior, the measured limit of the fast backend at its default order (Section 6.6) The oblique value is the one region where the fast backend is above 10^{-3} at its default order. Against the order-12/refine-3 continuum arbiter the fast backend is 1.6% at the closest moving in-support node, the arbiter-validated in-support accuracy, whereas the driver and the `exact` backend overestimate that node by 7.2× (Section 6.5).

The mesh alone, without the near shell, reproduces the field to 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ where the slices cut the bunch core (Section 6.5); these planes cut the sources, the region where the single-mesh spectral convolution has no near-field treatment. With the near shell enabled at the measured radii ($R = 3$ for $\gamma = 1$, $R = 4$ for $\gamma = 10$ in mesh nodes of the $h = 1$ slice grid), the assembly residual is at roundoff and the measured plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$ (Section 6.7); the residual is the mesh contribution outside the shell radius.

In the perpendicular direction at $\gamma = 10$, aspect 1, the measured $M-T$ error at $d = 48$ is 2.6×10^{-4}, the $M-P$ error is 1.5×10^{-2}, and the TSC smoothing $T-P$ is 1.48×10^{-2}, of which the $M-P$ error is the TSC smoothing, not the deposit (Table 8). At $\gamma = 100$, aspect 1, the perpendicular $M-T$ error at $d = 48$ is 3.75×10^{-1} and does not decrease with further increase in d At aspect 4, $\gamma = 1$, the perpendicular $M-T$ error at $d = 48$ is 2.4×10^{-3}. The axial direction is below 3×10^{-4} at $d = 6$ for every γ.

The γ-landscape measurement at $d = 8$ cells, $h = 1$, shows the error 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$. The h-refinement exponent p (error $\sim h^p$, fitted from $h = 1$ and $h = 1/4$) falls from 1.85 to about 1.0 as γ rises; the controlling length is the beaming cone width d/γ, and at $\gamma \geq 50$ it is sub-cell (Section 6.13). The detailed discussion is in Section 8.

6 Accuracy

Each error figure names its reference (M mesh, T continuum, P point). The three references stay apart throughout: the mesh M, the TSC-convolved continuum T (Eq. (1), the quantity the mesh is built to compute), and the point field P (the un-smoothed Liénard–Wiechert field). M against T is the discrete deposit and convolution error. T against P is the TSC smoothing, an $O(1)$ effect at $\gamma h \gg 1$. M against P is the total.

6.1 The reference and the metrics

The arbiter is an independent TSC-convolved continuum, a tensor Gauss–Legendre quadrature with a TSC-knot split and a singularity-cancelling power map. Its quadrature error is below the 3×10^{-4} gate in every configuration measured here.

Three metrics are reported and are not interchangeable, and each claim in this section names which one supports it.

Floored. $|\text{ref} - \text{mesh}| / \max(|\text{ref}|, |\text{mesh}|)$, over points where $|\text{ref}| \geq 10^{-3}$ of the local (six-component) field peak. This is the relative per-component error of record for the electric field.

Unfloored. The same ratio at every point. It is the only one that reaches the magnetic field, because the floor excludes it (below).

Absolute. $|\text{ref} - \text{mesh}| / \max_{\text{point}} |\text{ref}_c|$, the peak-normalized error; it supports whole-field statements.

The judge metric $|\text{ref} - \text{mesh}|/(1 + \max(|\text{ref}|, \text{mesh}|))$ is absolute-dominated: for fields of order 10^{-9} V/m the +1 denominator swamps the error, so it reports a crossover at $d = 2$ for every case and does not discriminate.

The floor hides the magnetic field. For $\gamma = 10$ every B point is excluded by the floor: $|\mathbf{B}|$ is about 10^{-10} of the E-dominated local field, far below the 10^{-3} threshold, so the floored metric reports no magnetic accuracy at all. The unfloored B relative error is 9.7×10^{-4} at $d = 32$ (single particle, aspect 1), against a B peak that the floored column never sees. The floored column of Table 9 is therefore electric-only at $\gamma = 10$.

6.2 Reference-separated error at 48 cells

Table 8 shows two things. At $\gamma = 10$, aspect 1, the mesh against the point field ($M-P = 1.5 \times 10^{-2}$) is about 58× the mesh against the TSC-convolved continuum ($M-T = 2.6 \times 10^{-4}$, an L2 ratio); the

difference is the TSC smoothing, which $M-P$ inherits and $M-T$ does not. A mesh-versus-point comparison measures the shape the deposit applies, not the discretization error. Second, the $M-T$ error is not converged perpendicular to the beam at $\gamma h \gg 1$: at aspect 1 it is acceptable only for $\gamma = 10$ and not for $\gamma = 100$; at aspect 4 it is not converged at any γ. Only the axial direction is clean at every γ.

6.3 The independent validation

A second, independent pass measured the shipped CLI end to end: `picBunch` on `rosi_v100`, against the order-12 independent continuum reference (refine 3), this work, $d = 2$ to 32 cells, $\gamma \in \{1, 10\}$, aspect 1 and 4, single particle and three-particle cloud.

Table 9 confirms the reference-separated picture independently. The measured value is 1.1×10^{-4} at $d = 12$ for $\gamma = 1$, aspect 1, and 2.6×10^{-4} at $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1 the relative error is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$. These are floored electric values; the unfloored magnetic error at $\gamma = 10$ is 9.7×10^{-4} at $d = 32$. The window starts at $d = 2$, so the TSC-support interior ($|s| < 1.5$ cells) is not covered by these sweeps; it is what the near shell measures (Sections 3.7, 6.8)

Fig. 3 plots the mesh against the TSC-convolved continuum; the three-particle cloud series is in Table 9. The series is floored, so the magnetic field, which the floor excludes at $\gamma = 10$ (Section 6), is not in the figure. The $\gamma = 10$ point at $d = 48$ (1.8×10^{-3}) is in Table 9 but outside the plotted window. The point-field column of Table 8 ($M-P$) is the mesh against the un-smoothed field, which carries the TSC smoothing in addition to the deposit error and never agrees with $M-T$ for $\gamma h \gg 1$.

6.4 Crossover radii

At $\gamma = 1$, aspect 1, the mesh is within 3×10^{-4} of both references from $d = 8$; the two are interchangeable there. From $\gamma = 10$ up, against the point field the measured perpendicular error at $d = 48$ is 6.5×10^{-3} ($\gamma = 10$, aspect 1); against the TSC-convolved continuum the perpendicular crossover is $d = 48$ ($\gamma = 10$, aspect 1), and for the remaining configurations the measured value at $d = 48$ exceeds the gate. The axial direction is 6 against the continuum at every γ.

6.5 Mid-bunch slices: field and error side by side

The distance sweeps above sample the far field. To measure the mesh in the bunch core we slice through it and compare, pixel by pixel, with an exact TSC-convolved reference: 64×64 planes, cell 10^{-7} m, $N = 200$, $Q = -100$ pC, $\gamma = 10$ with momentum along $+y$. Two cases share the positions and differ only in a 1.045 mrad rms transverse divergence in x (Figs. 4, 5, 6, 7); a separate 32×32 plane carries the binned 1 mrad case (Fig. 8). In every panel y is horizontal and increases left to right, so the bunch moves rightwards and y is the beam axis; the vertical axis is x in the x-y cut and z in the z-y cut. Each figure has a field row with four columns (E_x, E_y, E_z, E^2) and an error row underneath. The three components share one linear colour scale and bar; E^2 (or B^2) has its own scale and bar with its own unit label. The error row is logarithmic and masked where the reference is below 10^{-3} of its own peak. The forward case compares the shipped mesh plus fast shell against the order-4 adaptive Gauss–Legendre driver; that driver overestimates the worst moving oblique in-support point of this forward case by 7.2×, a localised failure at a specific node-particle configuration, so the solid white outline marks the TSC support and the in-support error pixels are shown against a reference that is itself unreliable at that node. The trustworthy in-support statement is the fast-versus-arbiter 1.6% at the worst node (Section 6.6). The fluctuation reference is complete, but picBunch v0.7.0 rejects a momentum spread, so only its reference field panels are shown and the picBunch panels are labelled unavailable.

Shell radius unit. The earlier $N = 1000$ slices run the default mesh with no near shell. The near-shell envelope measured in the crossover work, "3 to 4 PIC cells," assumed a mesh spacing $h = 1/4$ of the PIC cell, so $R = 3$ and $R = 4$ are 12 and 16 mesh nodes. On this grid the node spacing is $h = 1$ PIC cell (2×10^{-7} m), so the same $R = 3/4$ is $3/4$ mesh nodes, four times smaller in physical units than the envelope configuration. At $R = 3$ on an $h = 1$ grid the shell covers 3 mesh nodes; the TSC support extends ± 1.5 cells from each source, so the shell covers a fraction of the full support. The measured plane errors at this radius are reported in Section 6.7.

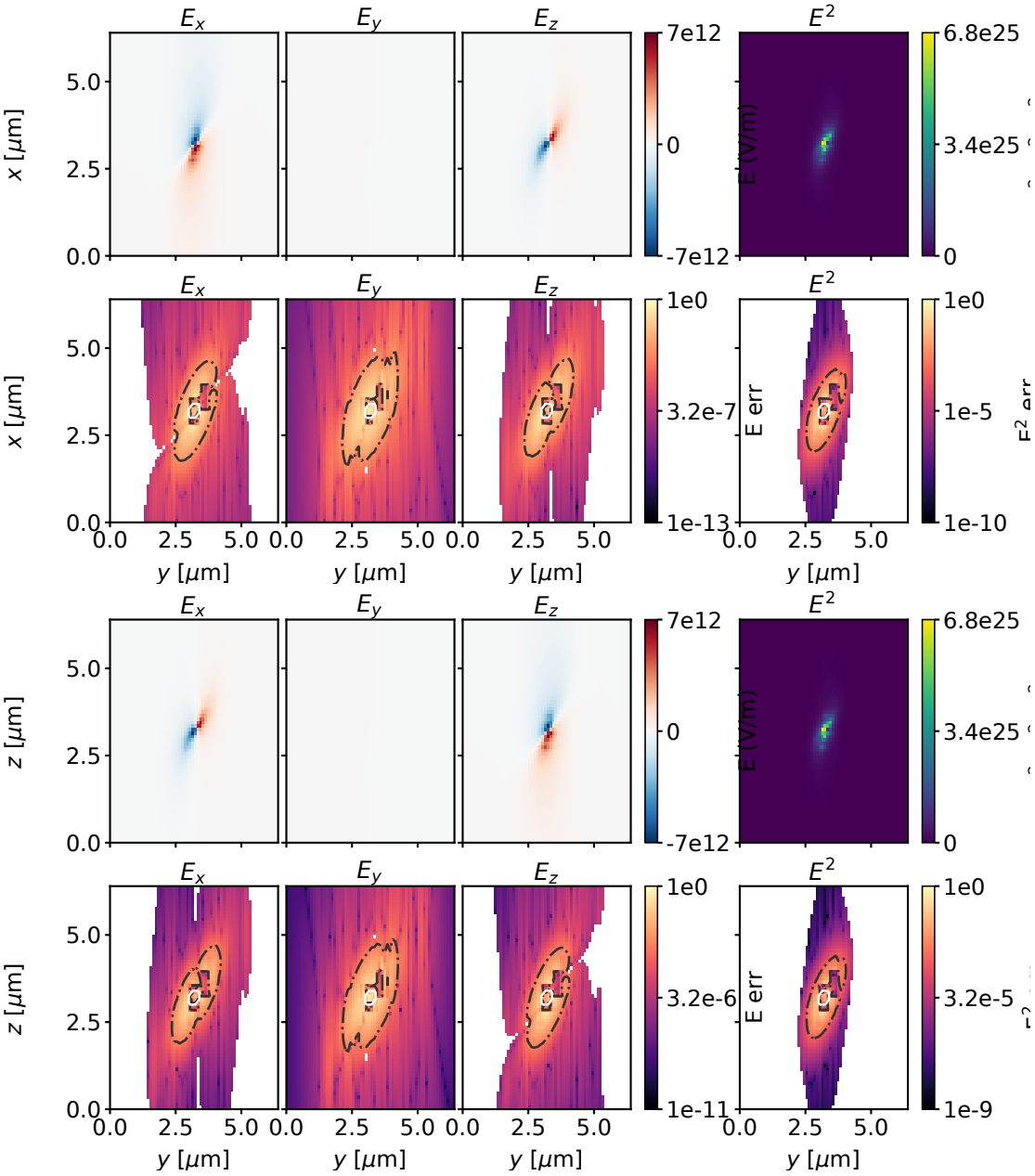


Figure 4: Forward-case electric validation ($N = 200$, $Q = -100\,\mathrm{pC}$, $\gamma = 10$ along $+y$, 64×64 planes, cell 10^{-7} m; x-y plane at $z = 3.2\,\mu\mathrm{m}$ (upper) and z-y plane at $x = 3.2\,\mu\mathrm{m}$ (lower), with y horizontal, the beam axis. Each image shows four columns E_x, E_y, E_z, E^2: the three components share one linear scale and bar, while E^2 has its own scale and unit label; the lower row is logarithmic error, masked below 10^{-3} of the peak, and the white outline marks the TSC support. The in-support fast-versus-arbiter error against the order-12 independent continuum reference (this work) is 1.6% at the worst node.

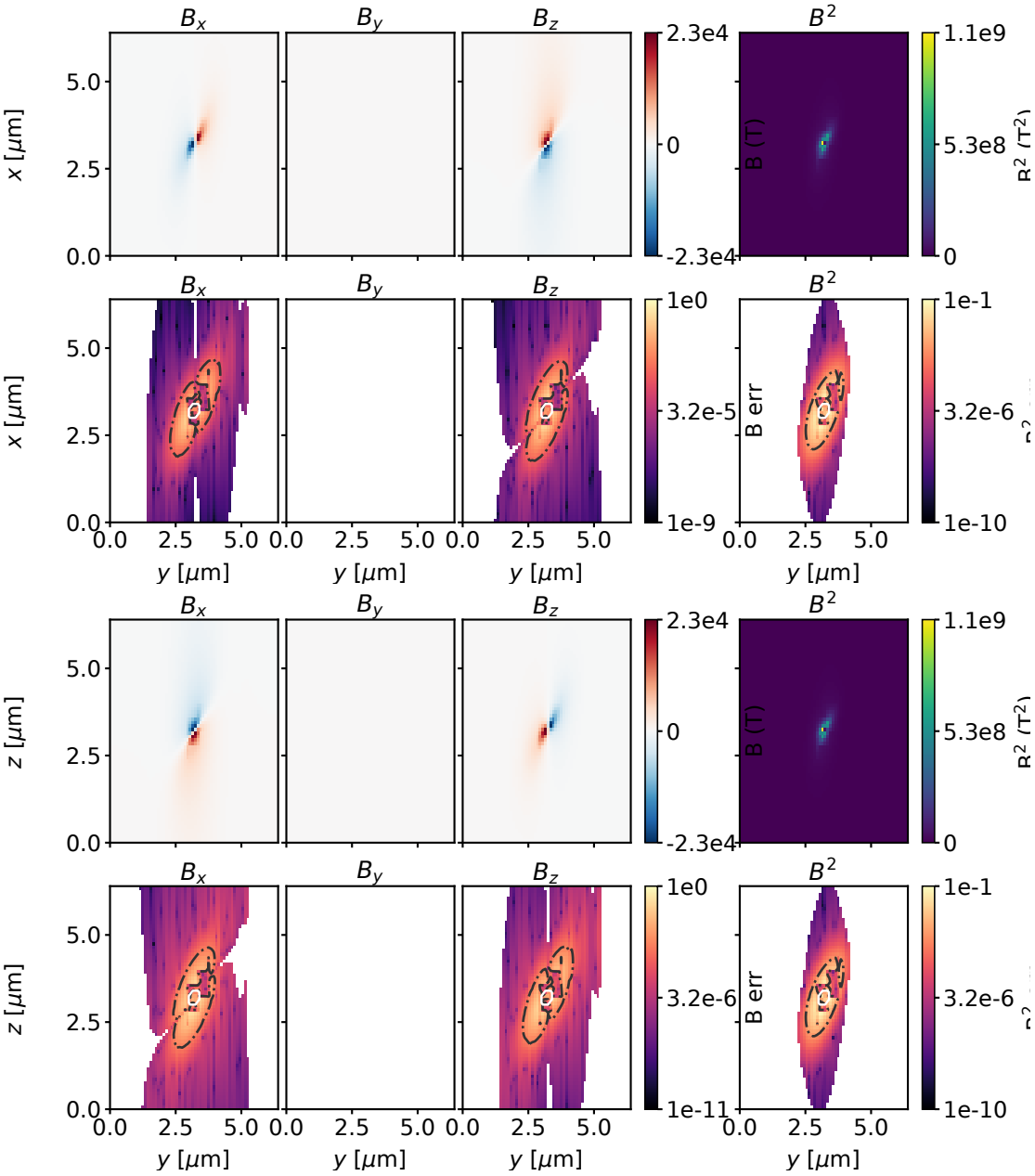


Figure 5: Magnetic companion to Fig. 4 with the same planes and layout, showing four columns B_x, B_y, B_z, B^2: the three components share one linear bar and B^2 has its own scale and unit label; $B_y \equiv 0$ for motion along $+y$ is marked in its panel. y is horizontal and is the beam axis, and the same TSC-support outline is drawn as in the electric figure. The magnetic field is about 10^{-10} of the E-dominated local field at $\gamma = 10$.

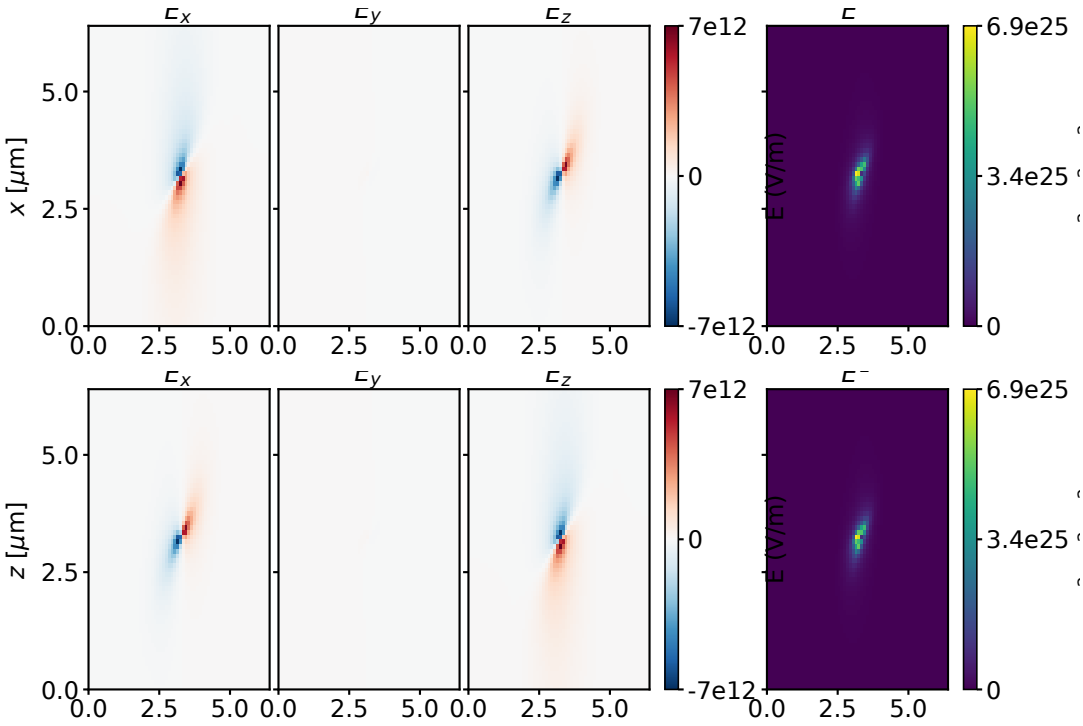


Figure 6: Fluctuation-case electric reference with the same geometry and bunch as Fig. 4 but a 1.045 mrad rms x-divergence, showing four columns E_x, E_y, E_z, E^2 from the order-12 independent continuum reference (this work): the three components share one linear scale and bar, while E^2 has its own scale and unit label. y is horizontal and is the beam axis. picBunch v0.7.0 rejects a momentum spread, so only the reference field is shown and the mesh panels are labelled unavailable.

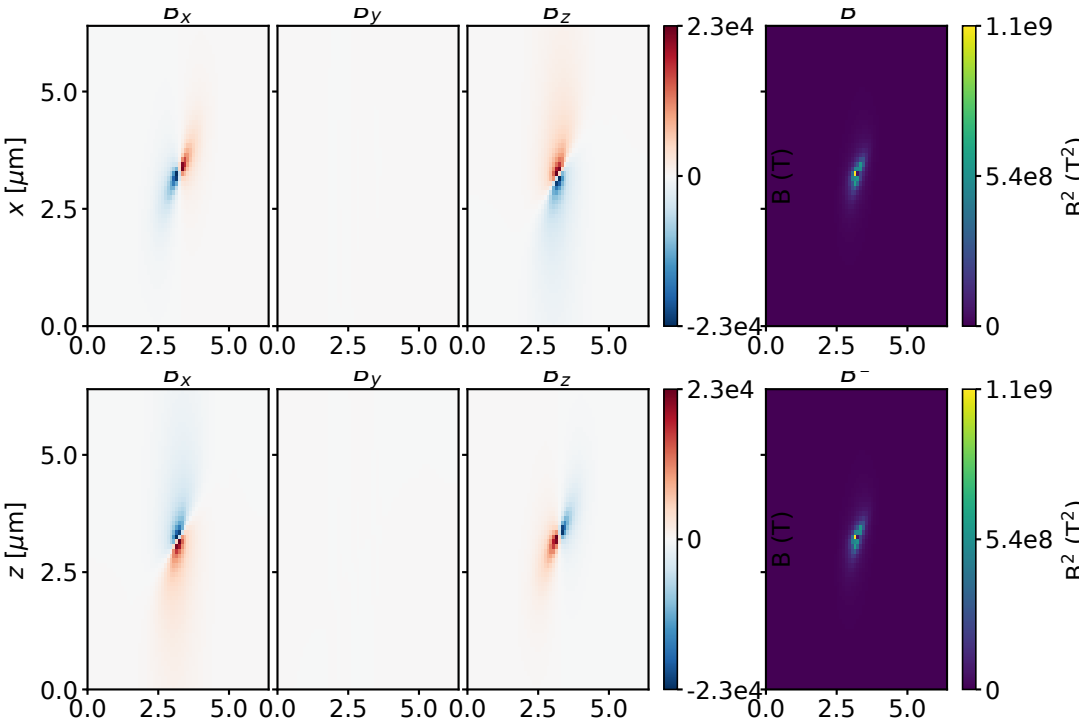


Figure 7: Magnetic reference for the fluctuation case, with the same planes, four columns B_x, B_y, B_z, B^2: the three components share one linear bar and B^2 has its own scale and unit label, as in Fig. 6; y is horizontal and is the beam axis. The x-divergence lifts B_y off zero, although it remains about 3×10^{-4} of the dominant component. Only reference fields from the order-12 independent continuum reference (this work) are shown; the picBunch panels are unavailable.

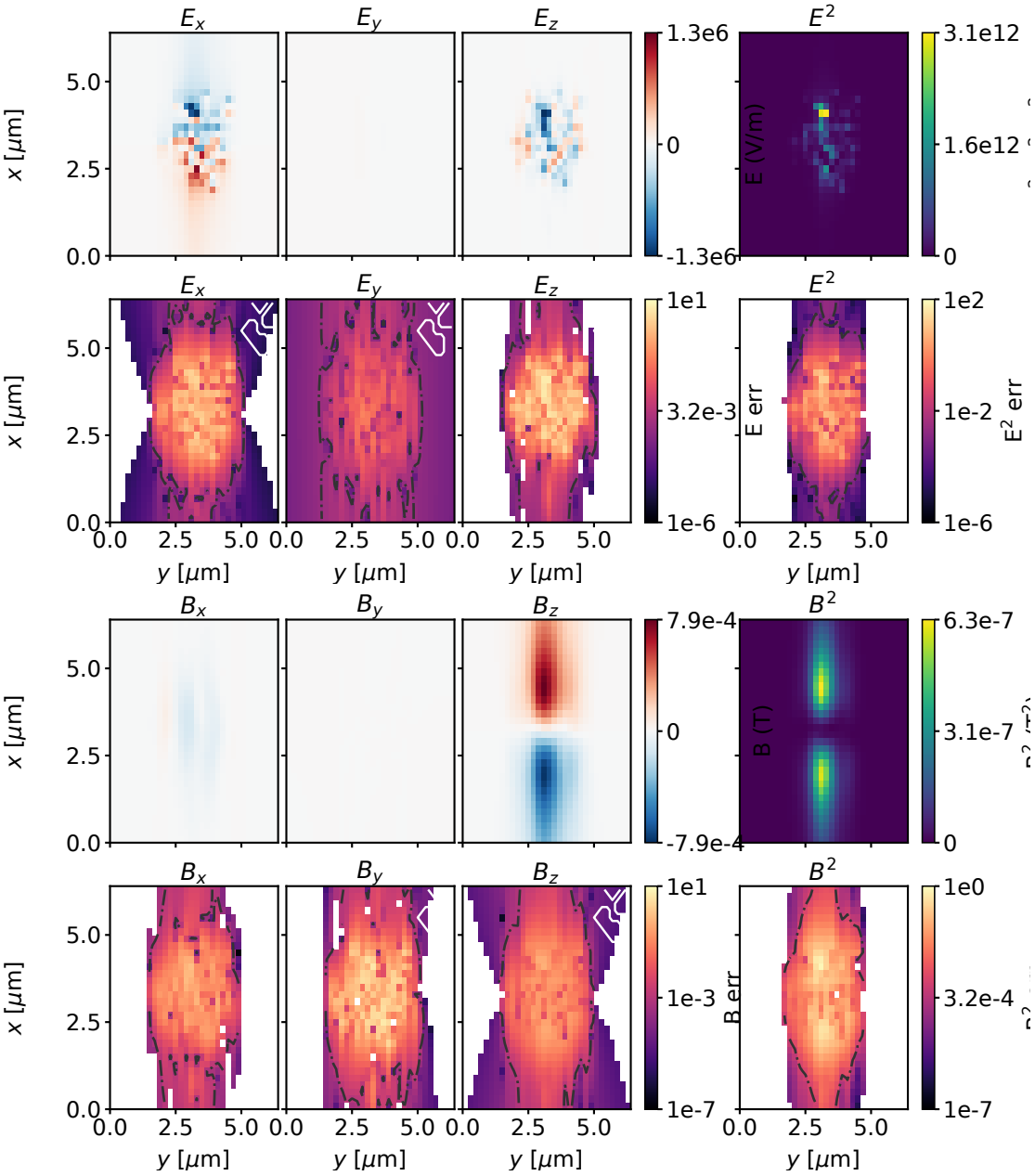


Figure 8: Binned 1 mrad case ($N = 200$, $Q = -32$ aC, $\gamma = 10$, 32×32 x-y plane at $z = 1.6$ µm, cell 2×10^{-7} m), with the picBunch velocity-binned field against the order-12 independent continuum reference (this work). The upper image is electric and the lower magnetic, each with four columns: three components on one shared scale and bar, and E^2 or B^2 on its own scale and bar; y is horizontal and is the beam axis, and error rows are logarithmic and masked below 10^{-3} of the reference peak. The solid white outline marks the TSC support.

Table 11 lists the measured worst relative error per component and the fraction of unmasked plane pixels above 10^{-3} for the earlier $N = 1000$, $32 \times 32 \times 16$ set, with error-column pixels below 10^{-3} of the plane peak masked white. The new validation figures above supersede those slices for the forward and the 1 mrad cases.

The error is largest where these planes cut the bunch core: the spectral far-field convolution is not the dominant term there, and the maps show the measured error concentrated at the beamed core and falling away toward the plane edges. The far field is measured separately (Table 9, Fig. 3); the near-shell contribution is measured at the same geometry in Section 6.7.

6.6 The fast near evaluator, per component, by region

Table 12 gives the worst per-component relative error of the fast evaluator against the independent continuum, by region and by γ. The metric is the artifact formula $\max|\text{mesh} - \text{ref}|/(1 + \max|\text{ref}|)$ over 20 interior, 16 face and 16 outside points, cells 10^{-7} m. "Interior" is inside the TSC support, "face" is on its boundary, and "outside" is 1.6 to 3.6 cells away. The stationary ($\gamma = 1$) case is order 5; the moving axial case is order 8; the moving oblique interior is order 5. The oblique value is the measured limit of the fast backend: the rest-frame singular box of the oblique direction measures 8.15×10^{-2} at its default order.

The per-component values behind the stationary interior row are 7.99×10^{-4}, 3.55×10^{-4}, 6.05×10^{-4}, 0, 0, 0 for $E_x, E_y, E_z, B_x, B_y, B_z$ (the magnetic field is identically zero for a stationary bunch); the moving axial interior row is 1.69×10^{-4}, 1.68×10^{-4}, 1.36×10^{-5}, 1.67×10^{-4}, 1.69×10^{-4}, 0; the moving oblique interior row is 2.43×10^{-2}, 7.42×10^{-2}, 6.32×10^{-2}, 8.15×10^{-2}, 2.96×10^{-2}, 3.89×10^{-2}

moving axial case meets 10^{-3} in every region; the moving oblique interior is 8.15×10^{-2}.

6.7 Near-shell accuracy and cost on the slice geometry

On the same $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m geometry, the exact near shell was run through the library path at the radius the envelope case assumed, $R = 3$ for $\gamma = 1$ and $R = 4$ for $\gamma = 10$, in mesh nodes of this $h = 1$ grid. The mesh-only plane errors (Table 11) are 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$: these planes cut the sources, the region where the spectral convolution is not the dominant term. With the near shell the worst per-component plane error is 1.6×10^{-3}–1.1×10^{-1} for $\gamma = 1$ (the high end is the z-y plane E_z, 0.1067) and 3.2×10^{-2}–5.7×10^{-1} for $\gamma = 10$: at $R = 3/4$ on an $h = 1$ grid the shell radius covers $3/4$ mesh nodes, which is a fraction of the TSC support extent (Section 6.5), and the measured residual is the mesh contribution outside the shell. The discrete Gauss-law indicator, the Yee backward-difference $\operatorname{div} E$ residual normalised to the peak deposited ρ/ε_0, moves from 0.3534 to 0.0895 for $\gamma = 1$ and from 5.026 to 0.399 for $\gamma = 10$, against reference values 0.0904 and 0.0814: $\gamma = 1$ reaches the reference value, $\gamma = 10$ drops 12.6× and stays 4.9× above it.

The assembly itself is not the limiting term: the tests of Section 3.7 show the tiled subtraction exact to roundoff, so the residuals above are the mesh outside the shell.

Runtime at $N = 1000$ is 4.3 min for $\gamma = 1$ and 59 min for $\gamma = 10$ (one 16-thread socket); the exact evaluator costs about 8× more per pair at $\gamma = 10$ than at $\gamma = 1$.

6.8 Cost at 2×10^6 particles, one H100

Table 13 gives the measured CLI wall times at 2×10^6 particles in a 20-cell bunch, grid $66 \times 68 \times 66$, cell 2.5×10^{-7} m, near-shell tile 4, on one `H100` (`jrc0880`, `sm_90`), with the achieved near error for each configuration. The near shell is the dominant term in every shell row; the mesh-only wall is 2.48 s, of which the compute is 0.116 s (deposit 34.8–40.5 ms, kernel 1.1 ms, FFT 7.3–7.6 ms, read-out 8.0–8.5 ms, writer 48.8–254.5 ms). The near evaluator is the shipped fast backend of Section 3.7, measured through the CLI, not as a standalone kernel. Pair counts are quoted in source-node pairs and in components: support 4.319×10^8 / 2.59×10^9, three-cell 1.458×10^9 / 8.75×10^9, four-cell 2.662×10^9 / 1.60×10^{10}, the neighbour counts of the 20-cell geometry.

The support-only shell costs 281.0 s end-to-end at 7.99×10^{-4} interior error, the three-cell shell 955.4 s, and the four-cell shell 1510.7 s, at the same interior error

$R = 1.5$ to $R = 4$ multiplies the end-to-end wall by 5.4× at fixed particle count and grid, while the accuracy of the in-shell region is already at 7.99×10^{-4} at $R = 1.5$. The shell rate on the integrated path is 1.55×10^6, 1.61×10^6 and 1.76×10^6 pairs/s (source-node pairs) for the three radii, about 2× slower per component than the standalone kernel, the overhead of the per-tile local FFT and launch in the integrated path Fig. 9 places the three evaluated regions (TSC support, near shell, mesh) around one source and shows the measured error against distance for the $\gamma = 1$ and $\gamma = 10$ axial cases.

6.9 Device backend for the fast near evaluator

`v0.7.0` carries a device backend, `-near-shell fast-gpu`: the same algebra as the host fast path, one thread per (pair, component), on the CUDA device. Against the host fast path it agrees to 5.4×10^{-16} to 8.0×10^{-16}; when no CUDA device is present the run falls back to the host path with a warning, so the flag is safe on host-only builds. The device backend is the fast path of Section 3.7 moved to the

device; the lid slice run, one near-shell tile of the shipped problem, is 14.13 s on the device versus 25.68 s on the host, a 1.82× speed-up for that stage

6.10 Distributed FFT

A distributed FFT decomposes the output field in z slabs, one slab per rank, and moves the z rows of each 2-D slice between ranks with `MPI_Alltoallv` transposes around each 2-D transform Against the replicated path it agrees to 2.2×10^{-16}, and the written outputs are bitwise equal at 1, 2 and 4 ranks, so the decomposition changes no value. At a 640^3 output the per-rank host memory is 41.1 GB at 2 ranks and 20.6 GB at 4, the quantity the decomposition buys: the output field is slab-local, while the bunch remains replicated on every rank. The transposes are 74–85% of the FFT-plus- transpose time, so the distributed path is a memory feature rather than a speed feature.

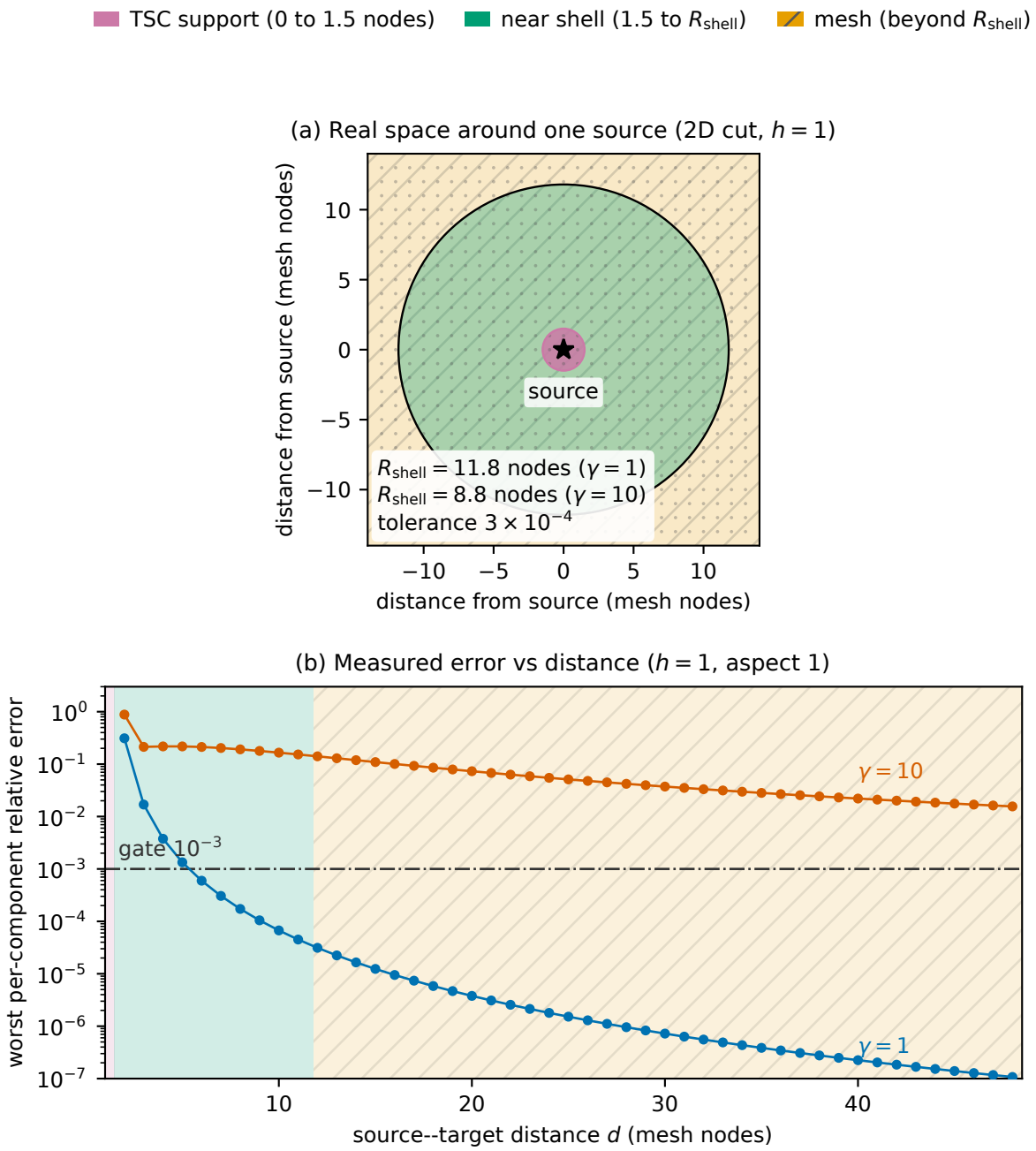


Figure 9: Where each method evaluates the field around a single source at $h = 1$ and aspect 1: TSC support to 1.5 nodes, the near shell out to R_{shell}, and the mesh beyond, with $R_{\text{shell}} = 11.8$ nodes for $\gamma = 1$ and 8.8 nodes for $\gamma = 10$ at the 3×10^{-4} tolerance. The lower panel plots the measured worst per-component error against distance for both γ, with the same regions shaded and the 10^{-3} gate marked. Field components sit at staggered Yee positions; panel (a) draws the in-plane offsets and labels the component pairs (for example E_x with B_y), while the out-of-plane E_z offset is not drawn.

6.11 The disagreement is per-source

On a $64 \times 64 \times 32$, cell 10^{-7} m, stationary bunch ($\gamma = 1$), the disagreement with the exact-quadrature reference does not grow with particle count: the band is flat in N from 1 to 1000 (10^{-3}–3×10^{-2}). The $O(1)$ values at $N \geq 10$ occur only because the global worst sample point falls 0.415–0.65 cells from a source, inside that source's TSC support; a charge-weighting defect would grow with N, and it does not.

6.12 Anisotropic cells: the gap is the TSC convolution

At anisotropic spacing the accuracy is quantified per geometry, not inherited from the cubic case. On a grid with per-axis cell spacing $(1, 1, 4)$ the stretched z axis matches the continuum to $\approx 10^{-3}$–10^{-7} across the in-grid distances, the same scale as the cubic control; the short x/y axes sit in the near-field TSC regime at a short physical distance ($d = 4$ cells is only 4 m along x/y, so $E_z \approx 0.19$ stationary and E_x/E_y up to ≈ 0.9 at $d = 4$ for the $\gamma = 10$ case). The mesh is not wrong because the spacings differ; the near points on the short axes are physically closer to the source.

The h^3 factor. The sampled convolution carries the Riemann volume of the continuum integral of Eq. (1): each sample covers h^3 cells, so the read-out applies $h_x h_y h_z/(M_x M_y M_z)$ (Section 3).

6.13 The γ landscape: 10 to 100

The error landscape for a single particle at $\gamma = 10, 50, 100$ is measured on a 96-cell box, cell 10^{-7} m, aspect 1, at $h = 1$ and $h = 1/4$, axial and oblique (45°), against the independent continuum (this work). The metric is the peak-normalised absolute error $\max|\text{mesh} - \text{ref}|/\max|\text{ref}|$ per component, at $d = 8$ cells for the h-comparison. These runs used padding 1 (padding 2 was infeasible for the 384^3 $h = 1/4$ grid); the $d = 48$ rows carry a small padding-1 periodic-wraparound floor of order 10^{-4} to 10^{-3}, so the $d = 48$ values are not padding-clean

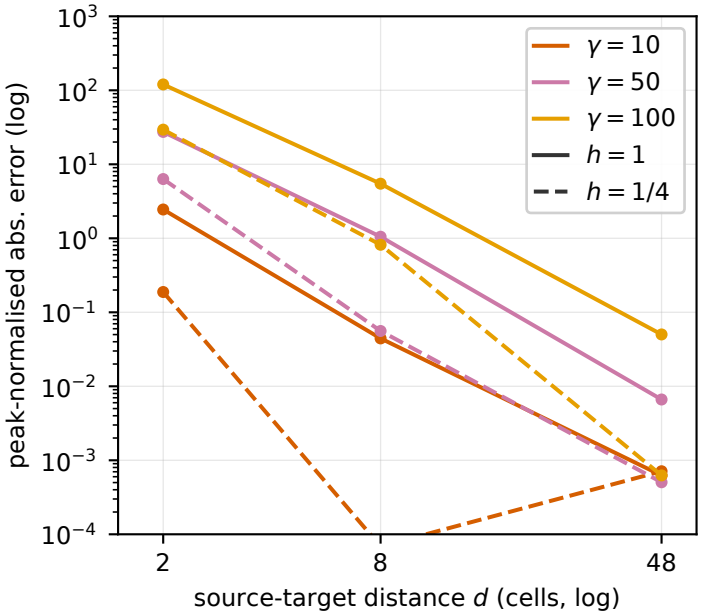


Figure 10: Peak-normalised absolute error against distance on log-log axes, single particle, 96-cell box, cell 10^{-7} m, aspect 1, axial, `jureca-h100`. Solid/dashed is $h = 1/1/4$ and hue is $\gamma = 10$ (vermillion), 50 (purple), 100 (orange). At $h = 1$ the error exceeds the peak field by $d = 8$ cells for $\gamma = 50$ and $\gamma = 100$.

Table 14 is the measured behaviour, and Fig. 10 shows the same quantities against distance. The error grows steeply with γ: 4.4×10^{-2} at $\gamma = 10$, 1.05 at $\gamma = 50$, 5.47 at $\gamma = 100$, at $h = 1$ and $d = 8$ cells (this work). Refining to $h = 1/4$ brings these to 6.9×10^{-5}, 5.6×10^{-2}, and 0.82. The controlling length is the beaming cone width d/γ: for $\gamma \geq 50$ the cone is sub-cell, the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) collapses to about 1, and refinement stops paying off. The h targets for 10^{-2} derive from the dominant $d = 2$ near-charge band (the error $\sim h^p$ fit uses the $d = 2$ band, not the $d = 8$ rows in Table 14). The oblique (45°) target is $h \leq 0.002$ at $\gamma = 50$; the axial (0°) target is $h \leq 0.001$ at $\gamma = 50$ (5.6×10^8 times the nodes). For $\gamma \geq 50$ the cone is sub-cell and 10^{-2} needs $h \leq 0.002$ ($\geq 10^8$ times the nodes); 10^{-3} is out of reach ($> 10^9$ to 10^{12} times the nodes) (this work). A shell of radius of order γ cells is what the near-charge error requires at $\gamma = 50$ to 100. These are measured limits of the mesh at a fixed distance in cells, stated as the behaviour the data show.

Table 1: Configurations of every measured case and the section that uses it (recorded values of this work; "not recorded" means no value was recorded). The two mid-slice entries give the bunch generation grid; the field grid for those cases ($32 \times 32 \times 16$ @ 2×10^{-7} m) is in the second block. Kinetic energy is $(\gamma - 1)\, m_e c^2$; the multi-velocity driver cell size is 10^{-6} m. The divergence $N = 200$ subset exists but its independent reference is pending, so it is not a row here.

case	N	total charge ($-e$ each)	size (µm; cells/axis)	spread	used in
mid slices, g10	1000	1.602×10^{-16} C	bunch generation grid $64 \times 64 \times 32$ @ 10^{-7} m (6.4 × 6.4 × 3.2 µm), $\sigma = 6$ cells/axis	none ($\gamma_{\min} = \gamma_{\max} = 10$)	§6.5, §6.7, §7
mid slices, g1	1000	1.602×10^{-16} C	bunch generation grid $64 \times 64 \times 32$ @ 10^{-7} m (6.4 × 6.4 × 3.2 µm), $\sigma = 6$ cells/axis	none ($\gamma_{\min} = \gamma_{\max} = 1$)	§6.5, §6.7, §7
large-domain, g10	100	1.602×10^{-17} C	extent 2.74 × 3.78 × 2.65	not recorded	§6 (Fig. 11)
large-domain, g1	100	1.602×10^{-17} C	extent 2.74 × 3.78 × 2.65	not recorded	§6 (Fig. 13)
cost, stationary	2×10^6	not recorded	20-cell bunch	none	§6.8
cost, moving axial	2×10^6	not recorded	20-cell bunch	none	§6.8
landscape, aspect 1	1	1.602×10^{-19} C	single particle, 96-cell box	none	§6.13
landscape, aspect 4	1	1.602×10^{-19} C	single particle at the origin	none	Table 8
multi-velocity, E 5%	512	not recorded	3-cell position cloud	5% energy (seed 100)	§6.16 (superseded)
multi-velocity, A 5°	512	not recorded	3-cell position cloud	5° angular (seed 103)	§6.16 (superseded)
multi-velocity, E 5% (v0.9.1)	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$	5% energy, $+y$ drift	§6.16
multi-velocity, A 5° (v0.9.1)	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$	5° angular, $+y$ drift	§6.16
1 mrad, binned (v0.9.1)	200	3.2×10^{-17} C	$\sigma = 6$ cells, centroid $(32, 32, 32)$, 1.046 mrad rms x-div	1 mrad, $+y$ drift	§6.16
divergence, 1 mrad	1000	1.602×10^{-16} C	$\sigma = 0.6$ each axis, centroid $(3.197, 3.208, 1.615)$	1 mrad paraxial, $+y$ (seed 43)	§6 (reference pending)

case	γ, kinetic E ($m_e c^2 = 510.999$ keV)	weight, cell	field grid (domain), host
mid slices, both	10 / 5.06 MeV; 1 / 0	1 each; 2×10^{-7} m	$32 \times 32 \times 16$ (6.4 × 6.4 × 3.2 µm), rosi_v100
large-domain, both	10 / 5.06 MeV; 1 / 0	not recorded; 2×10^{-7} m	$128 \times 128 \times 64$ (25.6 × 25.6 × 12.8 µm), jureca-h100
cost, both	1 / 0; 10 / 5.06 MeV	not recorded; 2.5×10^{-7} m	$66 \times 68 \times 66$, jureca-h100
landscape, aspect 1	10/50/100	1; 10^{-7} m	96^3 ($h = 1$), 384^3 ($h = 1/4$), jureca-h100 (ref: rosi_v100)
landscape, aspect 4	1/10/100	1; anisotropic, aspect 4	axis-aligned, $h = 1$, host not recorded
multi-velocity, both	10 / 5.06 MeV	not recorded; 10^{-6} m	48^3, jureca-h100
multi-velocity, both (v0.9.1)	10 / 5.06 MeV	1 each; 10^{-7} m	64^3, hal (A30, `sm_80`)
divergence	10 / 5.06 MeV	1 each; 2×10^{-7} m	$32 \times 32 \times 16$ (6.4 × 6.4 × 3.2 µm), rosi_v100

Table 4: Far-tail switch validation, 20-cell 4096-electron bunch, $R = 16.85$ cells, tolerance level 45.72 V/m (10^{-3} of the peak 45722.5 V/m). Worst per-component $|\mathbf{E}_{\text{tail}} - \mathbf{E}_{\text{direct}}|$ against an independent direct charge sum over 13 directions The switch at $6R$ is the one shipped: the error is 1.87×10^{-3} of the tolerance level.

switch radius	worst err (V/m)	/ tolerance level
$2R$ (33.7 cells)	4.80×10^{1}	1.05
$4R$ (67.4 cells)	8.77×10^{-1}	1.92×10^{-2}
$6R$ (101.1 cells)	8.53×10^{-2}	1.87×10^{-3}
$8R$ (134.8 cells)	1.66×10^{-2}	3.63×10^{-4}

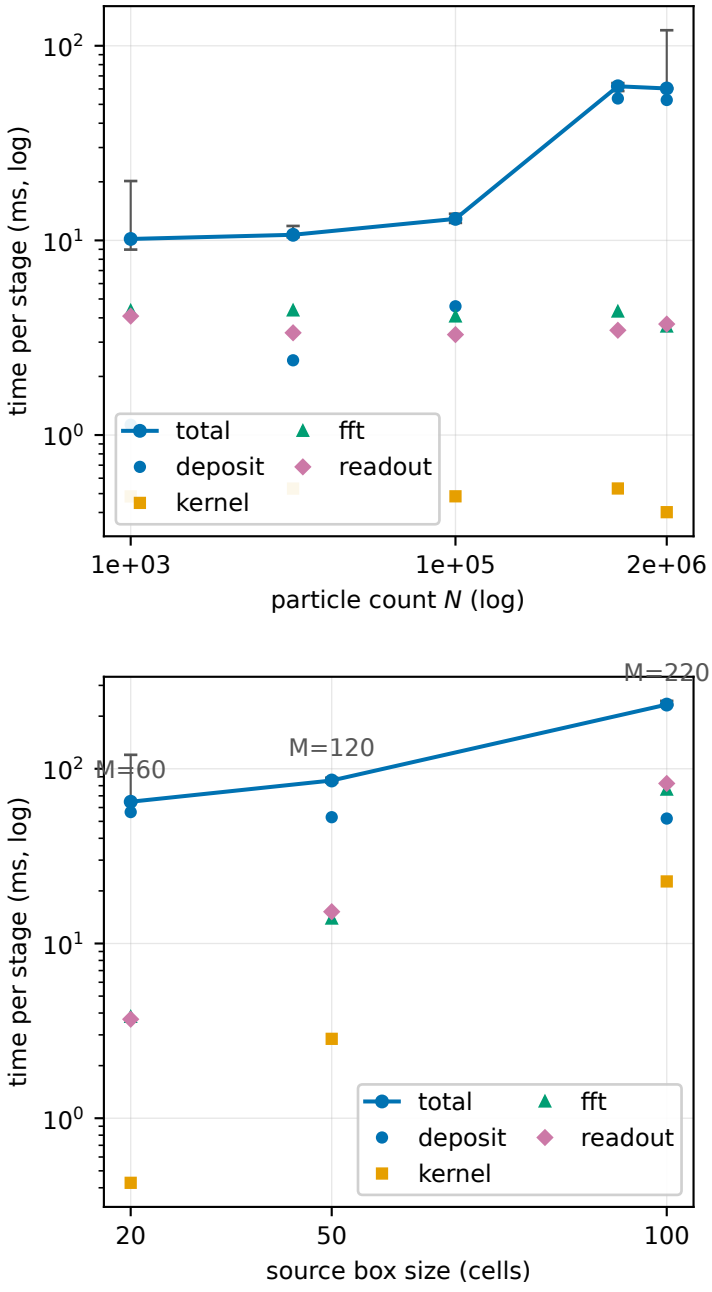


Figure 1: Pipeline time versus particle count (top, log-log) and source-box size (bottom, semi-log), padded array $M = 60/120/220$ cells, $\gamma = 10$, $h = 1$, double precision, `rosi_v100`/CUDA 12.6 (this work). Blue is the total and markers the per-stage medians; the deposit dominates at $N = 2 \times 10^6$ (57 of 64.8 ms), and the total rises from 64.8 to 233 ms as M grows from 60 to 220.

Table 5: Per-stage medians (ms) and device allocation (MiB), padding 2, double, `rosi_v100` GPU 0. "box" is the source-box size in cells; M is the padded FFT array per axis. Stage columns are medians over the three timed repeats.

N	box	γ	h	n	M	total	deposit	kernel	FFT	read-out	dev
10^3	20	10	1	30	60	10.17	1.13	0.48	4.36	4.08	8.4
10^4	20	10	1	30	60	10.69	2.42	0.53	4.37	3.35	8.7
10^5	20	10	1	30	60	12.90	4.59	0.48	4.07	3.28	11.4
10^6	20	10	1	30	60	61.92	53.61	0.53	4.31	3.45	38.9
$2{\cdot}10^6$	20	10	1	30	60	64.76	56.51	0.43	3.80	3.69	69.4
$2{\cdot}10^6$	50	10	1	60	120	85.67	52.82	2.85	13.89	15.20	127.4
$2{\cdot}10^6$	100	10	1	110	220	232.97	51.89	22.66	75.97	82.40	468.7
$2{\cdot}10^6$	20	10	1/2	50	100	69.27	52.38	1.29	7.43	8.16	99.5
$2{\cdot}10^6$	20	10	1/4	90	180	151.51	50.94	11.76	43.38	44.58	284.5

Table 6: End-to-end pipeline time including the checkpoint writer and the openPMD reader, medians over 3 repeats, `rosi_v100` GPU 0, 20-cell box, $\gamma = 10$, shipped CLI `-tolerance` 10^{-3} A different measurement path from the compute-only Table 5: the totals include the writer (the dominant added term); the per-stage split is deposit/kernel/FFT/readout/writer. The $N = 10^6$ total is 47.15 ms here versus 61.92 ms in the compute-only campaign.

N	total	deposit	kernel	FFT	readout	write
10^4	22.00	2.70	0.53	4.53	3.06	11.29
10^5	24.87	5.82	0.56	4.31	3.80	10.65
10^6	47.15	31.56	0.46	4.20	3.95	7.54

Table 7: Rank scaling on `rosi_v100`, idle window, medians of three Mesh: $N = 2{\times}10^6$, 112^3, component-parallel (replicated deposit and source FFT). Near shell: $N = 128$, 32^3, $R = 1.5$, tile 8, tile-parallel. Write: $N = 2 \times 10^6$, 256^3 global, 0.77 GB, collective HDF5. Mesh and near shell are bitwise identical at 1/2/4 ranks.

	near shell			mesh		write	
ranks	wall (s)	speedup	eff.	wall (ms)	speedup	wall (s)	speedup
1	36.61	1.00	100%	522.2	1.00	7.41	1.00
2	29.60	1.24	62%	436.3	1.20	4.34	1.71
4	27.29	1.34	34%	433.6	1.20	15.49	0.48

Table 8: L2-pooled relative error at $d = 48$ cells, $h = 1$, perpendicular direction (L2 over the six components; single-component worst: $\gamma = 1$, aspect 4, $M{-}T$ 2.65×10^{-5}; $\gamma = 10$, aspect 1, 1.4×10^{-3}). $M{-}T$: mesh vs TSC-convolved continuum; $M{-}P$: mesh vs point field; $T{-}P$: TSC smoothing, mesh-independent. At $\gamma = 10$, aspect 1 the $M{-}P$ vs $M{-}T$ gap is the smoothing (L2 ratio 58×), not the mesh.

γ	aspect	$M{-}T$	$M{-}P$	$T{-}P$
1	1	1.38×10^{-7}	9.89×10^{-8}	3.96×10^{-8}
1	4	2.65×10^{-5}	2.39×10^{-3}	2.40×10^{-3}
10	1	2.6×10^{-4}	1.5×10^{-2}	1.48×10^{-2}
10	4	3.33×10^{-2}	1.19×10^{-1}	1.44×10^{-1}
100	1	3.75×10^{-1}	2.20×10^{-1}	4.40×10^{-1}
100	4	2.78×10^{0}	2.49×10^{-1}	8.32×10^{-1}

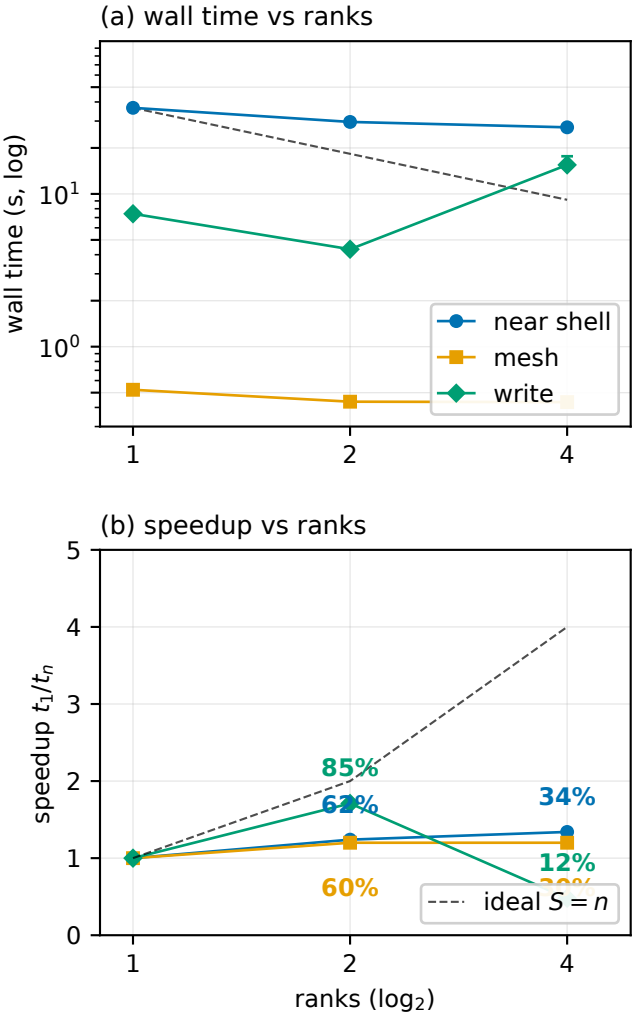


Figure 2: Strong scaling of the three rank-parallel pieces on `rosi_v100` (idle window, medians of three runs) for 1–4 ranks, with dashed ideal $1/n$ and $S = n$ references. The near shell reaches 1.34×, the component-parallel mesh 1.20×, and the collective writer 1.71× at two ranks then 0.48× at four.

Table 9: Worst per-component relative error (over x, y, $|\text{ref}| \geq 10^{-3}$ of the local field) of the shipped CLI against the TSC-convolved continuum, at $d = 16$ and $d = 32$ cells; rosi_v100, the order-12 independent continuum reference (refine 3), this work. "s/c" = single / cloud. Measured values at $d = 48$: $\gamma = 10$, 1.8×10^{-3} ($d = 32$: 6.5×10^{-3}); $\gamma = 1$ is below 3×10^{-4} from $d = 12$ (aspect 1) and $d = 32$ (aspect 4).

γ, aspect	s/c	d=16	d=32
1, 1	3.5e−5 / 1.0e−4	2.2e−6 / 2.0e−5	
1, 4	2.8e−3 / 2.1e−3	2.0e−4 / 2.6e−4	
10, 1	5.6e−2 / 5.3e−2	6.3e−3 / 6.5e−3	
10, 4	5.5e−2 / 5.0e−2	6.3e−3 / 6.4e−3	

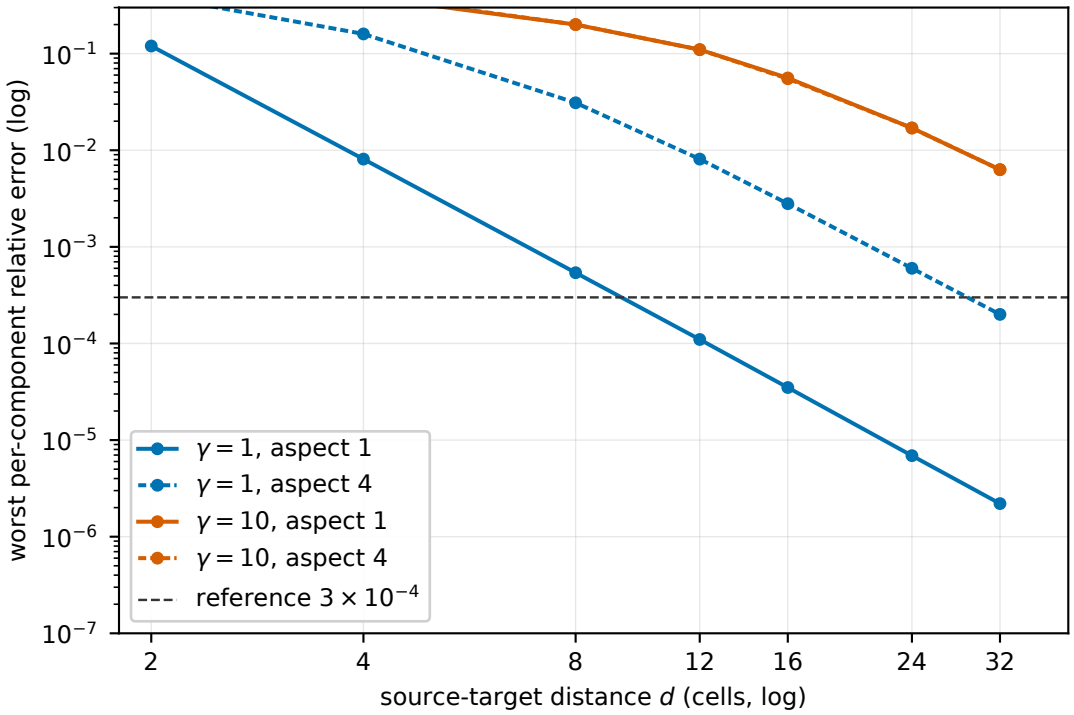


Figure 3: Worst per-component relative error against source-target distance d for the single-particle floored electric series on `rosi_v100`, both axes logarithmic; hue is γ (blue $\gamma = 1$, vermillion $\gamma = 10$) and line style is aspect (solid 1, dashed 4), against the order-12 independent continuum reference (refine 3), this work. The error falls from 1.2×10^{-1} at $d = 2$ to 2.2×10^{-6} at $d = 32$ for $\gamma = 1$, aspect 1.

Table 10: Crossover radius in cells: first d at which every larger d is at or below the gate, $h = 1$, L2 over the six components, per axis direction, worst over the three positive axis directions (>48: beyond the measured window), this work. T: TSC-convolved continuum; P: point field. Oblique directions are $\sim 1.5\times$ larger. The $\gamma = 1$ aspect 1 row is $d = 8$ against both references here, versus the $d = 12$ of the shipped-CLI series in Table 9: the two use different component sets (six axis directions, L2-pooled, versus per-component over x, y) and different implementations (this work).

			3×10^{-4}		10^{-3}	
γ	aspect	ref	axial z	perp x	axial z	perp x
1	1	T	8	8	6	6
	1	P	8	8	6	6
1	4	T	6	24	6	16
	4	P	48	> 48	32	> 48
10	1	T	6	48	6	48
	1	P	> 48	> 48	32	> 48
10	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48
100	1	T	6	> 48	6	> 48
	1	P	> 48	> 48	32	> 48
100	4	T	6	> 48	6	> 48
	4	P	> 48	> 48	32	> 48

Table 11: Mid-bunch slice statistics, mesh-only. Worst relative error per component and fraction of unmasked plane pixels above 10^{-3}, x-y plane at $z = 1.60\,\mu\text{m}$ and z-y plane at $x = 3.20\,\mu\text{m}$, $N = 1000$, $32 \times 32 \times 16$, cell $2 \times 10^{-7}\,\text{m}$, mesh against the independent exact TSC driver (FP64 quadrature, order 4, max level 5, tolerance 10^{-3}). The $\gamma = 10$ bunch moves along $+y$; y is the horizontal axis in every map. Node spacing $h = 1$ PIC cell; no near shell. The $\gamma = 1$ reference uses the archived tolerance 10^{-4} (driver floor 9.075×10^{-4}, 14 not converged).

plane	component	worst rel. err	frac $> 10^{-3}$
$\gamma = 10$			
x-y ($z = 1.60\,\mu\text{m}$)	E_x	1.81	54.1%
	E_y	1.16×10^{-1}	49.9%
	E_z	5.33	78.3%
	B_x	6.49×10^{-1}	72.7%
	B_z	3.77×10^{-1}	58.7%
z-y ($x = 3.20\,\mu\text{m}$)	E_x	5.06	90.4%
	E_y	1.17×10^{-1}	61.1%
	E_z	1.92	82.4%
	B_x	4.34×10^{-1}	80.9%
	B_z	7.16×10^{-1}	84.3%
$\gamma = 1$ *(stationary)*			
x-y ($z = 1.60\,\mu\text{m}$)	E_x	1.20×10^{-1}	22.8%
	E_y	1.34×10^{-1}	23.3%
	E_z	4.15×10^{-1}	74.7%
z-y ($x = 3.20\,\mu\text{m}$)	E_x	4.15×10^{-1}	49.6%
	E_y	1.37×10^{-1}	61.7%
	E_z	1.23×10^{-1}	53.9%

Table 12: Fast near evaluator, worst per-component relative error against the order-12 independent continuum reference, cells $10^{-7}\,\text{m}$, `jureca-h100` (this work). Regions: interior (inside the TSC support), face (on its boundary), outside (1.6–3.6 cells). The "remainder order" column is the Gauss–Legendre order of the smooth remainder; the singular `asinh` box is O(1) in every row (moving axial uses the γ-stretched box, moving oblique the rest-frame octant box at order 8). Stationary and oblique remainder order 5, moving axial 8. The moving oblique interior is the measured limit of the fast backend.

γ, direction	remainder order	interior	face	outside
1, stationary	5	7.99×10^{-4}	1.96×10^{-5}	1.57×10^{-7}
10, axial	8	1.69×10^{-4}	1.11×10^{-4}	2.12×10^{-6}
10, oblique	5	8.15×10^{-2}	3.94×10^{-3}	1.01×10^{-4}

Table 13: Measured CLI wall at 2×10^6 particles, 20-cell bunch, $66 \times 68 \times 66$, cell $2.5 \times 10^{-7}\,\text{m}$, tile 4, one `H100` (`sm_90`). "near-shell wall" excludes the mesh, "end-to-end wall" includes it. The mesh-only compute inside its 2.48 s wall is 0.116 s; the per-stage split (deposit, kernel, FFT, read-out, writer) is in the text.

configuration	near-shell wall (s)	end-to-end wall (s)	worst near error
mesh only	0	2.48	no near correction
support-only shell, $R = 1.5$, order 5	278.28	281.0	7.99×10^{-4}
three-cell shell, $R = 3$, order 5	908.14	955.4	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out
four-cell shell, $R = 4$, order 5	1508.12	1510.7	7.99×10^{-4} in, $\leq 1.6 \times 10^{-7}$ out

Table 14: Single-particle mesh error at $d = 8$ cells, 96-cell box, cell $10^{-7}\,\text{m}$, aspect 1, axial, `jureca-h100`. Peak-normalised absolute error, worst per component, at $h = 1$ and $h = 1/4$. The error grows with γ; at $h = 1$ the $\gamma = 50$ and $\gamma = 100$ values exceed the peak field, and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises.

γ	$h = 1$	$h = 1/4$	slope p ($h = 1 \to h = 1/4$)	note
10	4.4×10^{-2}	6.9×10^{-5}	1.85	
50	1.05	5.6×10^{-2}	1.06	$h = 1$ off by more than the peak
100	5.47	0.82	1.01	$h = 1$ off by more than the peak

Fig. 11 shows the $\gamma = 10$ field and error on a $128 \times 128 \times 64$ domain, cell 2×10^{-7} m, $N = 100$, against the independent continuum at tolerance 10^{-3}; Fig. 12 is its magnetic companion, and Fig. 13 is the stationary-bunch electric case. The boundary field is about 10.3% of the peak (worst per-component, $\gamma = 10$ z-y E_z), so the colour scale is set by the bunch and the far decay is visible inside the frame. The core error is the method error of the mesh where the planes cut the bunch, far above the 10^{-3} reference floor.

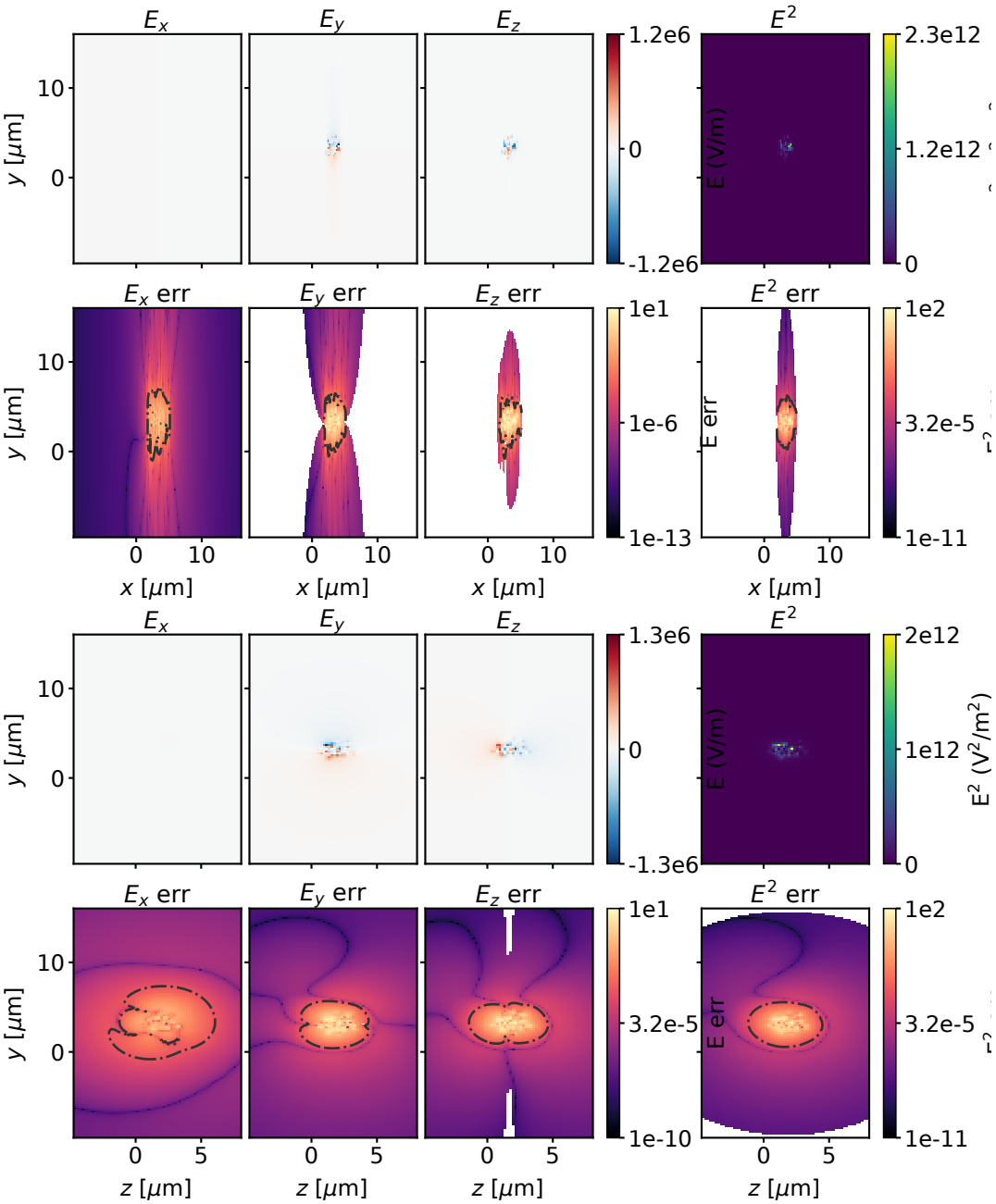


Figure 11: Large-domain electric slices ($\gamma = 10$, $128 \times 128 \times 64$, cell 2×10^{-7} m, $N = 100$, `jureca-h100`) from the earlier axial-x data set (not the $+y$ beam-axis set); x is horizontal in the x-y cut and is the beam axis there, while z is horizontal in the z-y cut. Upper and lower images are x-y and z-y planes, each showing four columns E_x, E_y, E_z, E^2: the three components share one linear scale and bar, and E^2 has its own scale and unit label, above a logarithmic error row against the order-4 driver reference. The boundary field is 10.3% of the peak (worst component, z-y E_z), so the colour scale is set by the bunch.

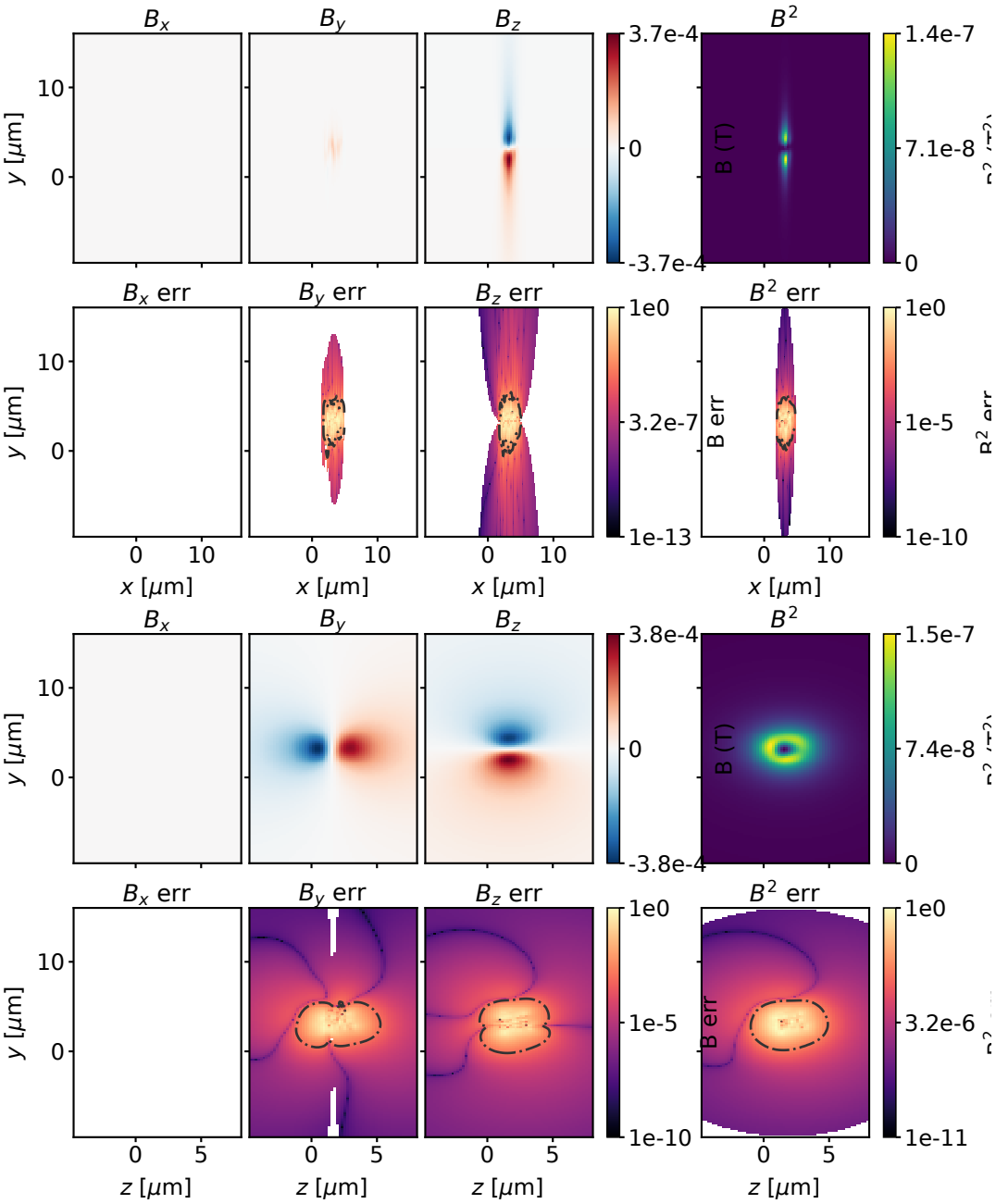


Figure 12: Magnetic companion to Fig. 11 at $\gamma = 10$, same planes and earlier axial-x geometry; x is horizontal in the x-y cut and is the beam axis there, while z is horizontal in the z-y cut, and $B_x \equiv 0$ is marked in its panel. Each image shows four columns B_x, B_y, B_z, B^2: the three components share one linear bar and B^2 has its own scale and unit label, with the same logarithmic error convention as the electric figure. The plane peak is 6.2×10^{-4} T.

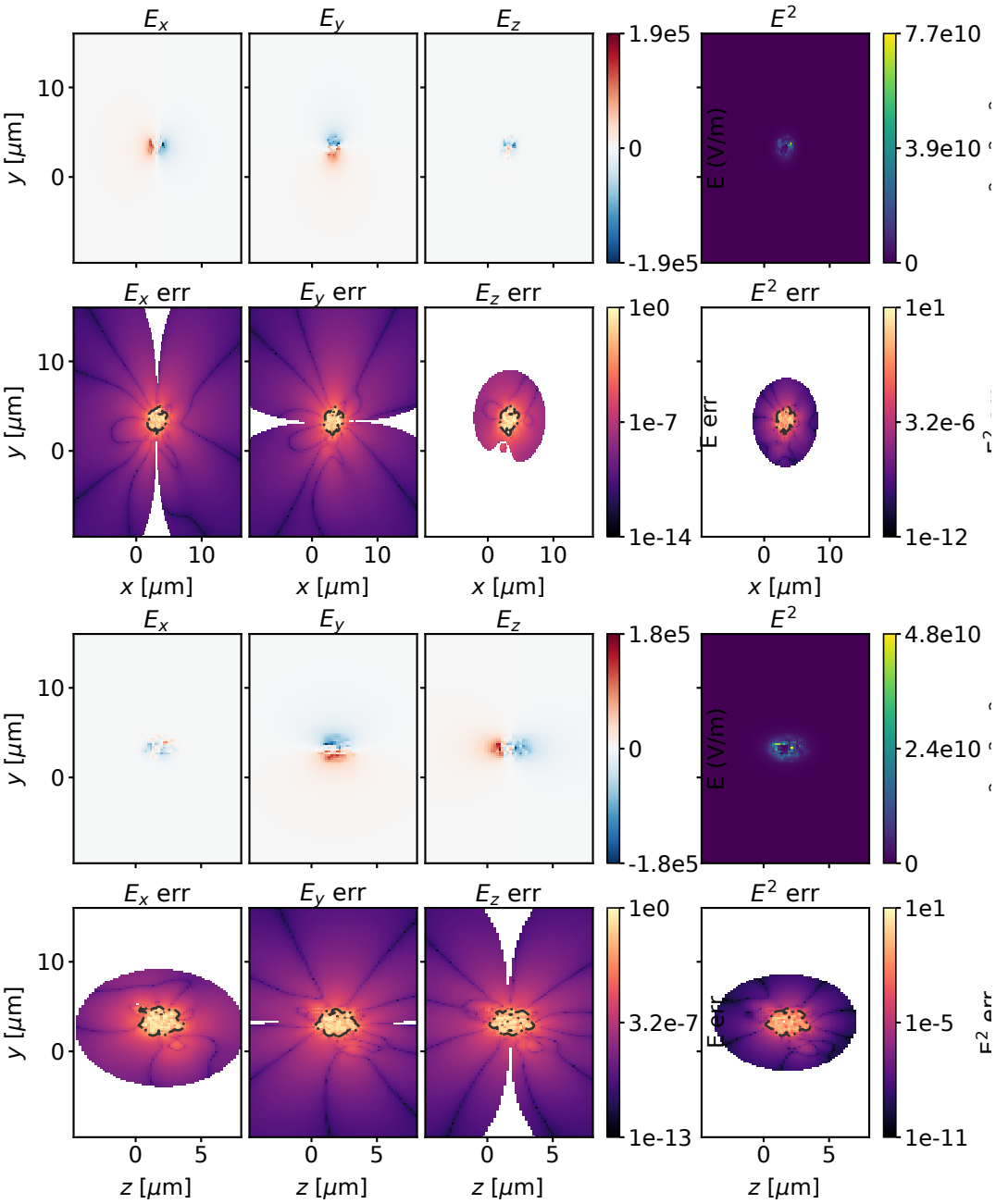


Figure 13: Large-domain electric slices for the stationary bunch ($\gamma = 1$), same domain and layout as Fig. 11; x is horizontal in the x-y cut and z is horizontal in the z-y cut (the bunch is stationary, so there is no beam axis). Each image shows four columns E_x, E_y, E_z, E^2: the three components share one linear bar and E^2 has its own scale and unit label, above a logarithmic error row. The core error reaches 7.6×10^{-1} (z-y E_x), far above the 10^{-3} reference floor.

6.14 Single-particle validation at seven velocities ($\gamma = 1.0$, 1.1, 1.5, 2, 10, 50, 100)

Figures 14–20 show the shipped mesh field for one electron on the final 128^3 grid, cell 5×10^{-8} m over the same 6.4 μm domain, against the independent order-12 continuum reference on the mid-z x-y plane. The seven velocities are $\gamma = 1.0$, 1.1, 1.5, 2, 10, 50 and 100. Each figure is one velocity case, with y horizontal and increasing left to right, and its rows are labelled with γ, β and the particle count $N = 1$. Table 15 lists the measured worst relative errors: 3.474×10^{-3}, 3.669×10^{-3}, 3.719×10^{-3}, 5.098×10^{-3}, 1.197×10^{-1}, 1.426 and 3.213 in velocity order. The error grows with γ because the field is compressed into a fraction of a cell: at $\gamma = 50$ and $\gamma = 100$ the Lorentz pancake is thinner than the mesh spacing,

Table 15: Seven-velocity single-particle validation on the final 128^3 grid, cell 5×10^{-8} m, 6.4 μm domain, against the order-12 continuum reference. Worst is the maximum relative error over the six components on the mid-z x-y plane.

γ	β	worst relative error
1.0	0	3.474×10^{-3}
1.1	0.4166	3.669×10^{-3}
1.5	0.7454	3.719×10^{-3}
2.0	0.8660	5.098×10^{-3}
10	0.9950	1.197×10^{-1}
50	0.99980	1.426
100	0.99995	3.213

so those cases show the geometric sub-cell limit of the mesh, not a solver defect. Raising the resolution from 64^3 to 128^3 at the same physical domain left the relative errors unchanged, so the growth with γ is the field compression rather than a discretisation that finer cells remove. The $\sim 3.5 \times 10^{-3}$ floor at low γ is the reference/quadrature floor at the settings used, not the solver's limit. The peak squared fields scale as $1/\text{cell}^2$ compared with the 64^3 series (a factor 16 for the halved cell), a consistency check on the two campaigns.

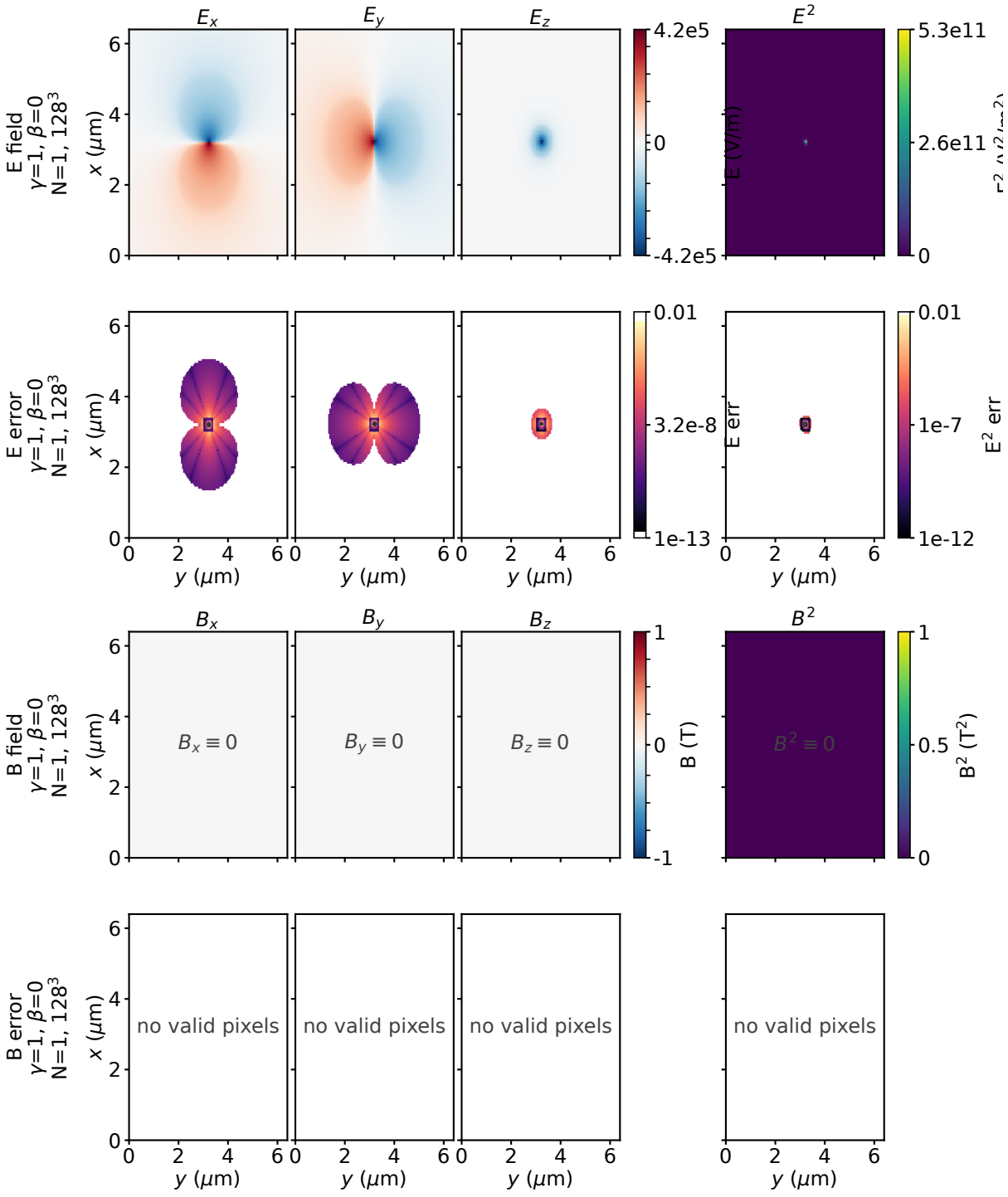


Figure 14: Single-electron field and error on the x-y plane at mid-z for $\beta = 0$ ($\gamma = 1.0$) on the 128^3 grid, cell 5×10^{-8} m. The E block and the B block each put E_x, E_y, E_z (or B_x, B_y, B_z) on one shared colour scale with its own bar and E^2 (or B^2) on a separate colour scale with its own bar, so the squared-magnitude scale never dominates the component scale. Error rows are logarithmic, carry the 10^{-3} contour, and are masked below 10^{-3} of the reference peak; at $\gamma = 1.0$ the B error bars are omitted because no pixel exceeds the mask threshold; the worst component error is 3.474×10^{-3}.

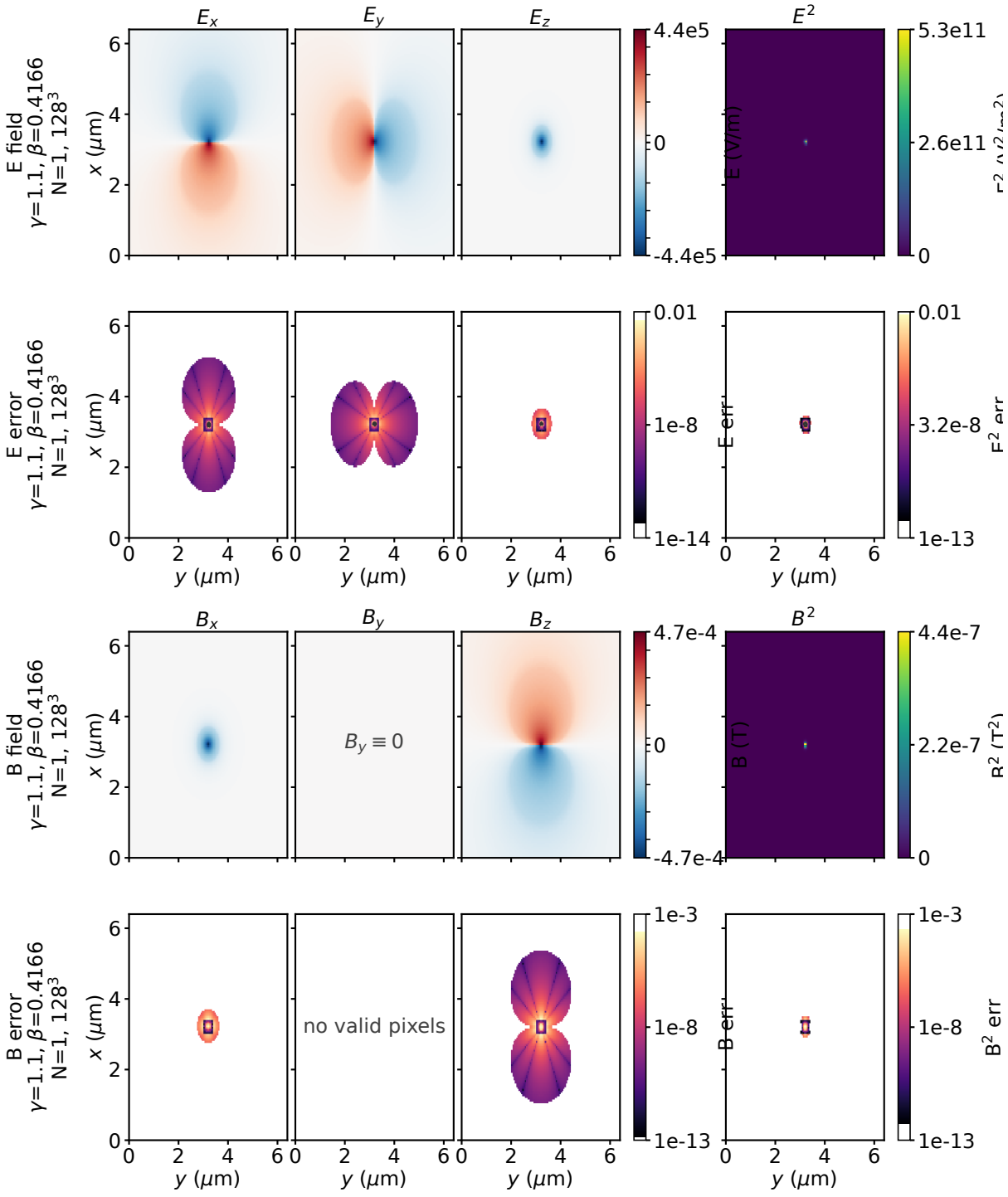


Figure 15: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.4166$ ($\gamma = 1.1$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 14. The worst component error is 3.669×10^{-3}.

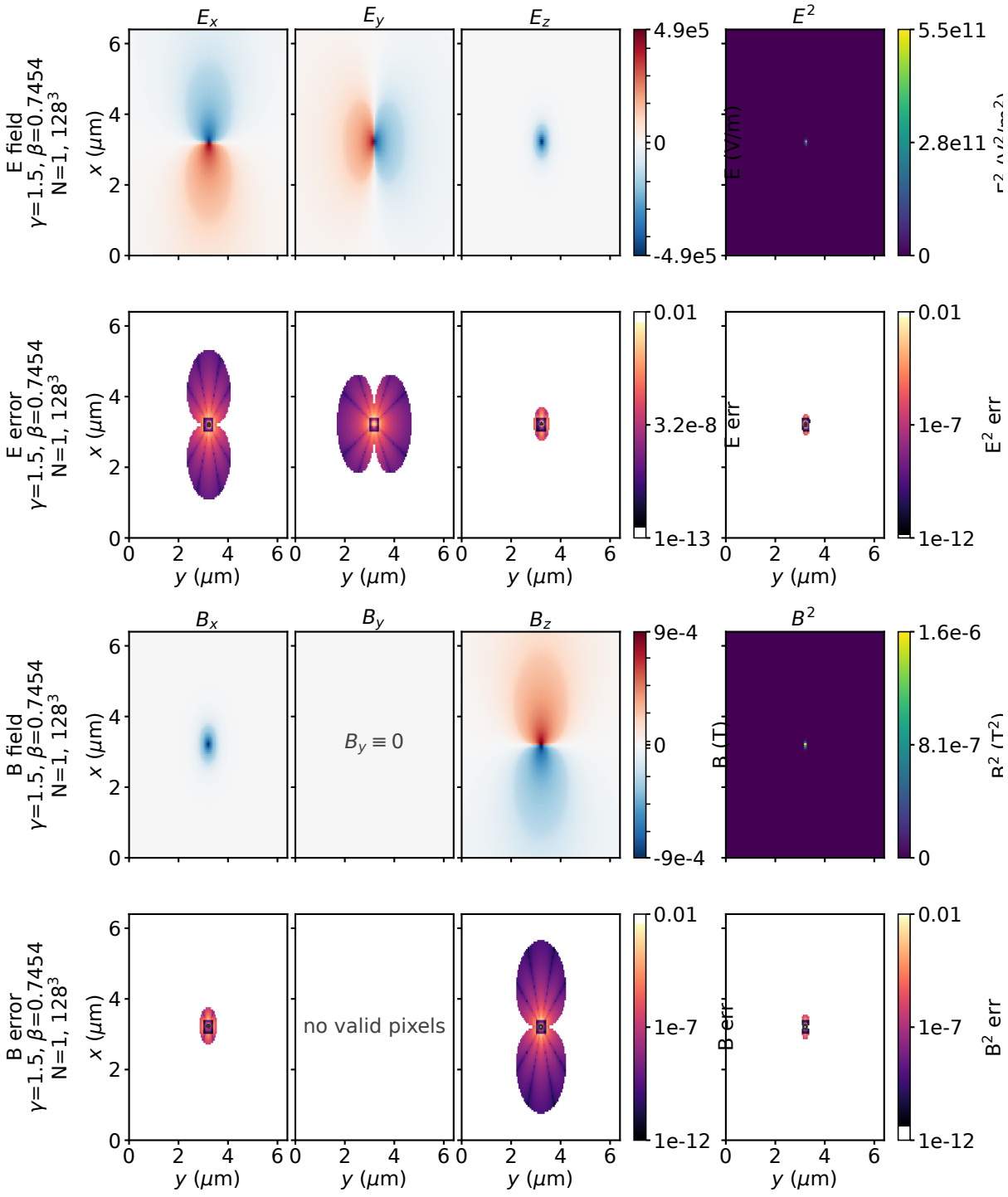


Figure 16: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.7454$ ($\gamma = 1.5$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 14. The worst component error is 3.719×10^{-3}.

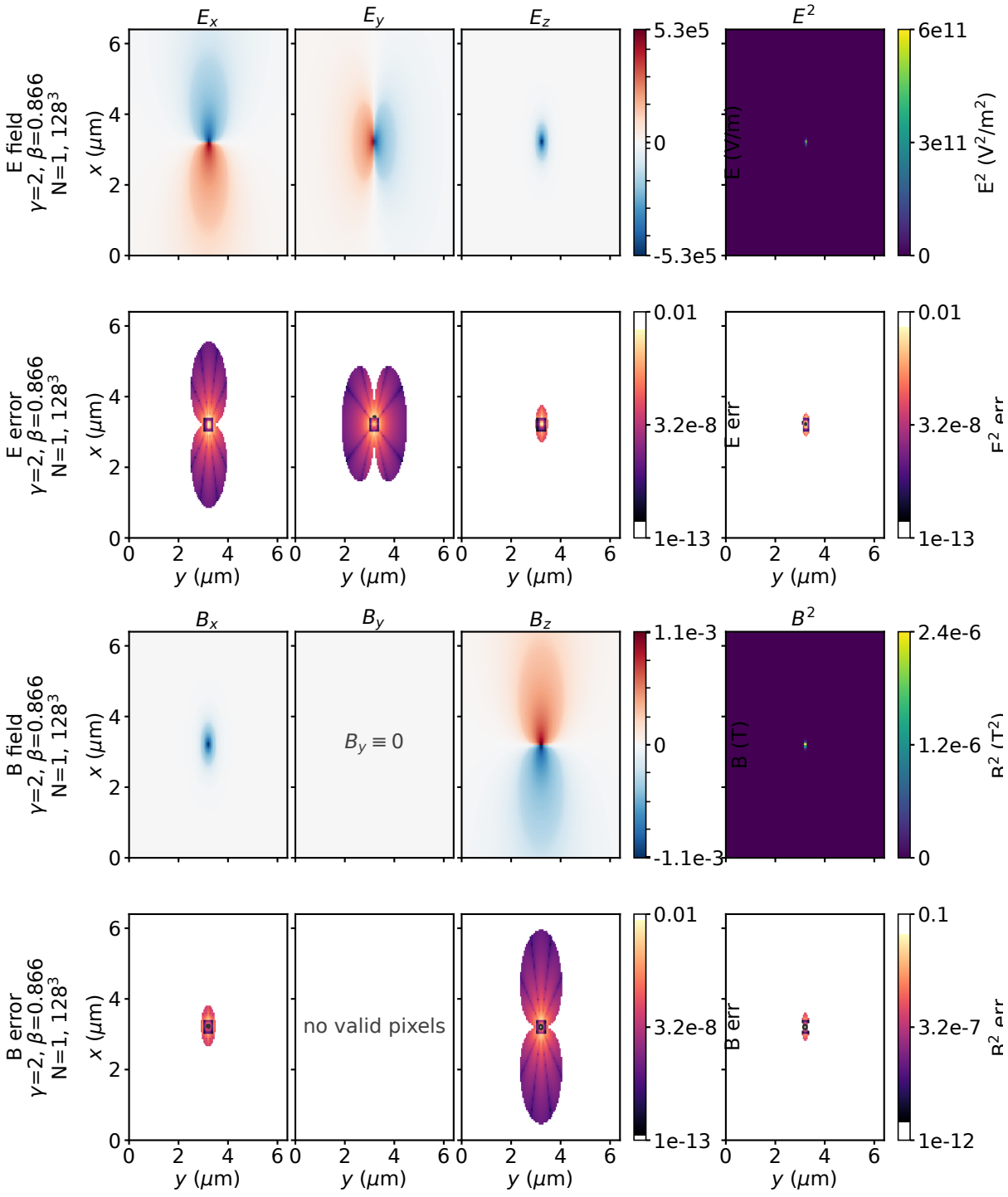


Figure 17: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.8660$ ($\gamma = 2.0$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 14. The worst component error is 5.098×10^{-3}.

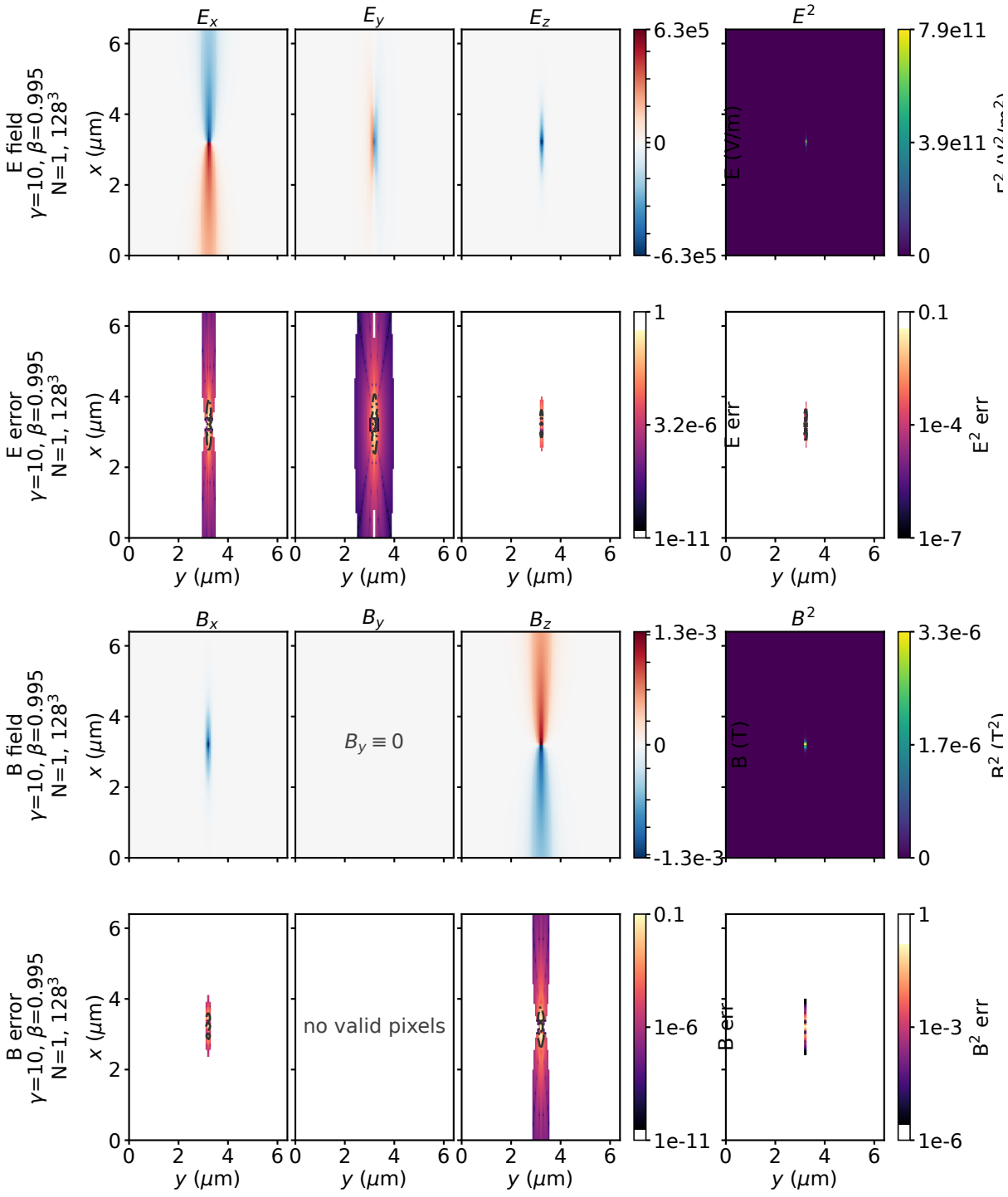


Figure 18: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.9950$ ($\gamma = 10$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 14. The worst component error is 1.197×10^{-1}.

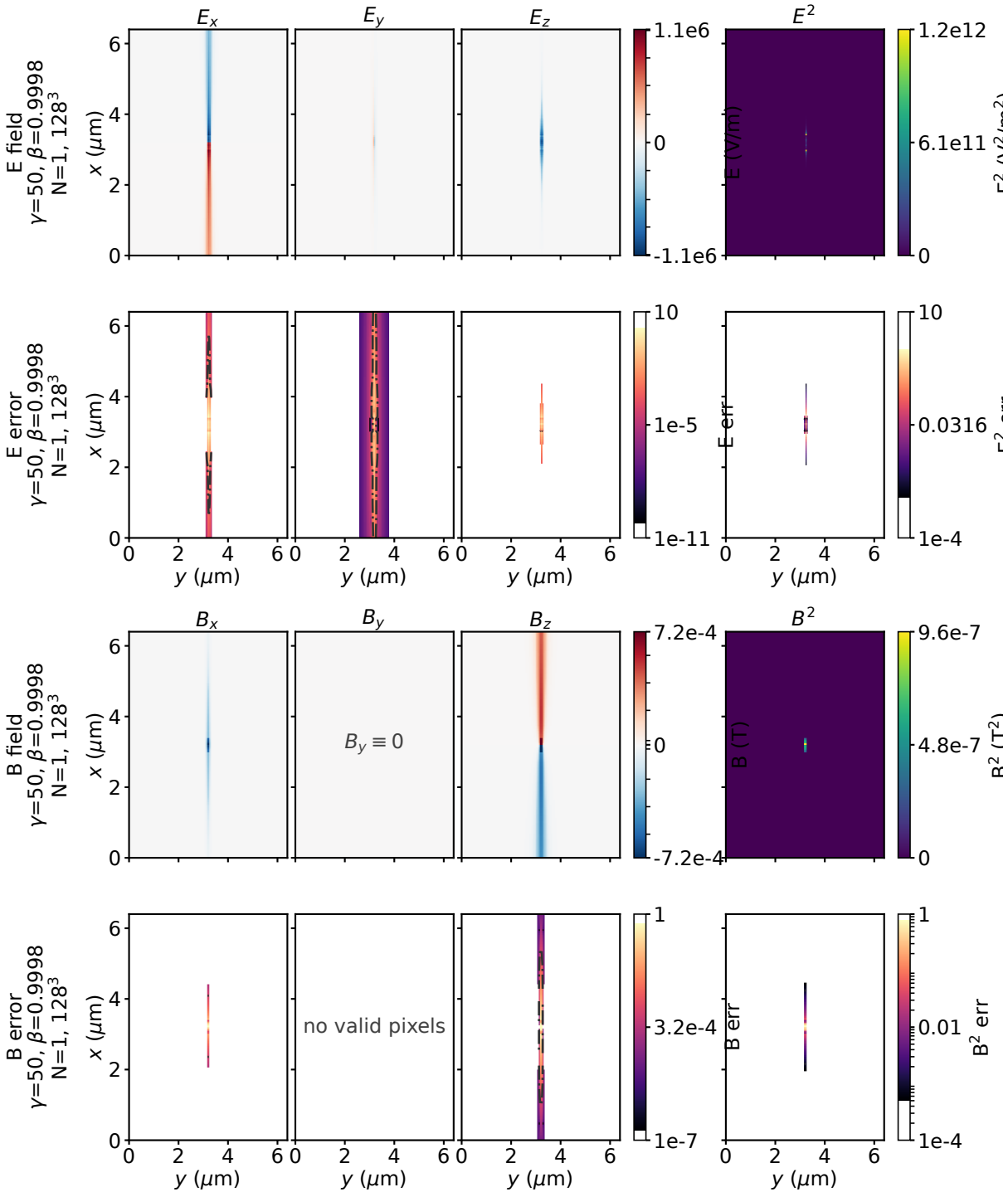


Figure 19: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.99980$ ($\gamma = 50$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 14. The worst component error is 1.426.

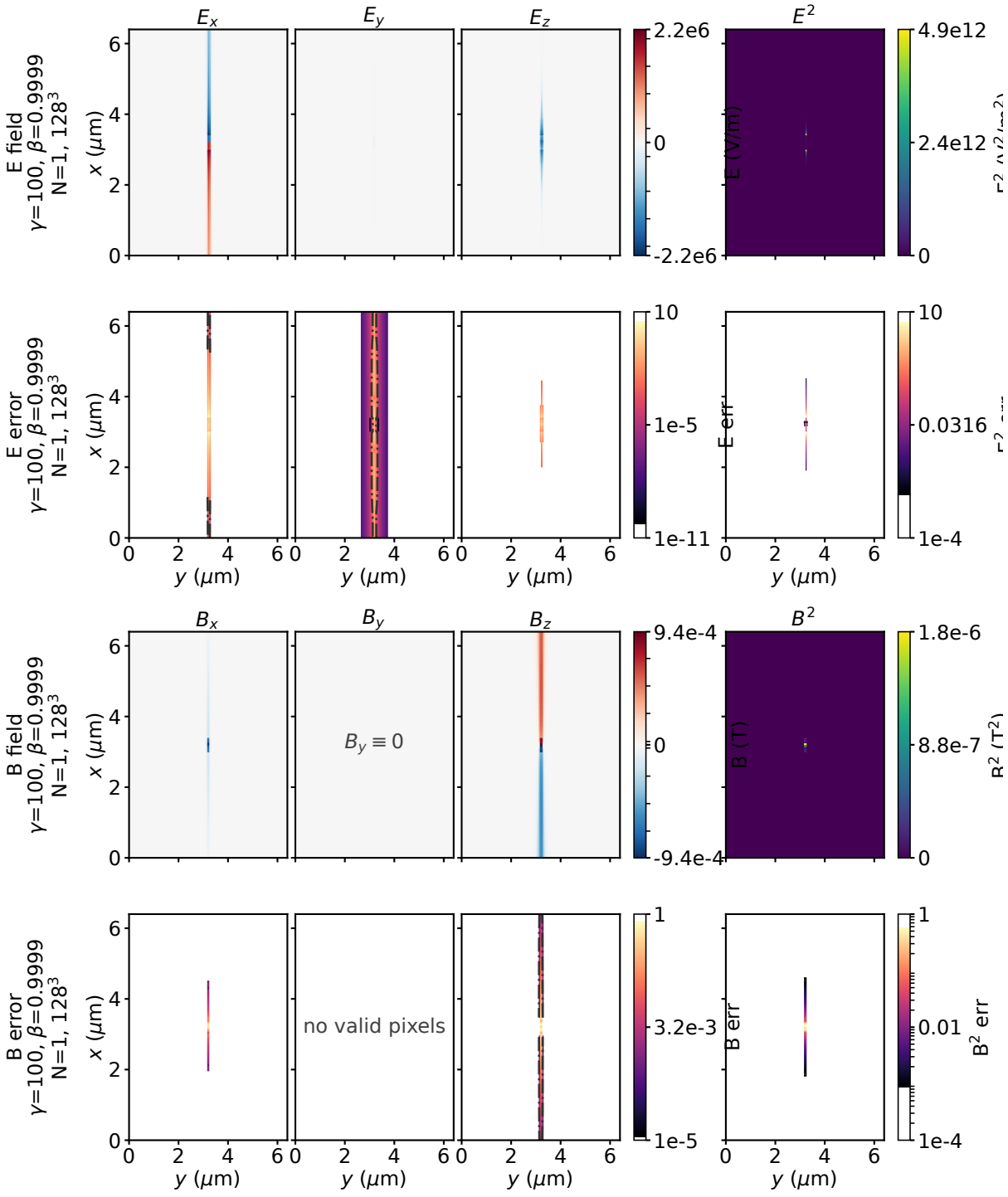


Figure 20: Single-electron field and error on the x-y plane at mid-z for $\beta = 0.99995$ ($\gamma = 100$) on the 128^3 grid, cell 5×10^{-8} m, with the same field, colour-bar and error conventions as Fig. 14. The worst component error is 3.213.

6.15 Determinism

Repeated runs of the shipped CLI are bit-identical: two runs of the same input produce byte-identical E/B mesh payloads (md5 `d3641d64...`, 15 092 736 bytes). The CLI also rejects a below-floor `-tolerance` (10^{-15}) with a nonzero exit. The `gridUnitSI` is applied once, in the reader (`OpenPMD.cpp`), and the mesh geometry follows `gridSpacing` · `gridUnitSI`; a mesh without a positive `gridUnitSI` is rejected.

Table 16: Binned multi-velocity peaks, 64^3 grid, 200 sources, $\gamma = 10$, picBunch `v0.9.1`.

spread	B=16 $\lvert B_z\rvert$ (T)	B=64 $\lvert B_z\rvert$ (T)	B=16 $\lvert B_y\rvert$ (T)	B=64 $\lvert B_y\rvert$ (T)
energy 5%	2.32050×10^4	2.32049×10^4	0	0
angular 5°	2.48202×10^4	2.48213×10^4	1.2948×10^3	1.2955×10^3

6.16 Binned multi-velocity binning, measured

The single-velocity scope of Section 2 is extended by splitting the velocity distribution into B classes and running one convolution per class; the near shell needs no binning, because each pair carries its own velocity. The measurements below are on a 64^3 grid with 200 sources on `hal` (NVIDIA A30, `sm_80`, CUDA 13.4.92), picBunch `v0.9.1`.

Single-velocity sanity. With zero spread the binned path is bitwise equal to the unbinned path: $B = 1$ and $B = 16$ give bit-identical fields in all six components ($\max|\Delta| = 0$).

Figure 21 shows the sampled distributions and the post-fix bin comparison. With a 5% energy spread and $+y$ drift the B_z peak is 2.32050×10^4 T at 16 bins and 2.32049×10^4 T at 64 bins, 0.0003% apart; $B_y = 0$ exactly, which is correct for a purely longitudinal current. With a 5° angular spread the B_z peak moves from 2.48202×10^4 to 2.48213×10^4 T and B_y from 1294.8 to 1295.5 T; the ratio B_y/B_z is 5.22×10^{-2}, and the two bin counts agree to 0.0043% in B_z and 0.056% in B_y.

1 mrad case against the arbiter. The fluctuation bunch ($N = 200$, $\gamma = 10$, $+y$ drift, 1.046 mrad rms x-divergence,) was re-run at 64 bins and compared against the Gauss–Legendre arbiter (unbinned, per-particle u, 698-node slice subset) measured before the fix. On the x-y slice ($z = 32$) picBunch gives $B_y = 7.835$ T and $B_z = 2.3336 \times 10^4$ T, against arbiter values of 7.759 T and 2.344×10^4 T (factors 1.010× and 0.996×); on the z-y slice ($x = 32$) the corresponding values are 7.561 T / 7.519 T and 2.3159×10^4 T / 2.327×10^4 T (1.006× and 0.995×). The 3-D B_y peak is 8.734 T and B_z is 2.3336×10^4 T. The worst component deviation is 1.4% (x-y ratio); every component on every slice is within 1.0%. Before the fix the same 64-bin run gave $B_y = 1.196 \times 10^3$ T, 154× the arbiter, and $B_z = 4.002 \times 10^4$ T, 1.71× the arbiter.

FFT buffer defect (fixed in `v0.9.1`). The replicated mesh path allocated its five transform buffers with 3-D `allocUnified` extents. On CUDA and HIP that allocation pads every innermost row to 128 B when the padded M_x is not a multiple of 16 doubles. The per-bin `memset` is pitched: it clears the row data but not the gap bytes, which lie inside the flat prefix the contiguous FFT plan and the flat deposit, kernel, and read-out loops read as data. The velocity-bin loop reuses the buffers, so the second and later non-empty bins read the previous bin's charge from the gaps, and the field grew with the bin count while the deposited charge summed correctly. The scope is exactly that case: more than one non-empty velocity bin on CUDA or HIP with a padded M_x that is not a multiple of 16. A single-velocity run clears a fresh buffer once and is unaffected. `v0.9.1` allocates all five buffers as flat 1-D counts. The `picBunchFftSizeCheck` tool (one allocation, a pitched `memset` and a forward/inverse pass per bin, summed against the direct free-space reference) confirms the fix: at the two failing local-box sizes $M = 24$ and $M = 40$ the free-space relative error is 3.5×10^{-16} to 7.0×10^{-16} at every bin count, versus 91.6 and 48.9 with the old 3-D buffers at 128 bins (this work). Single-velocity fields are bitwise unchanged between the v0.9.0 and v0.9.1 builds (SHA-256 verified on the 20-cell mesh-only case and the accuracy anchor), so no previously published single-velocity result moves.

Runtime: $B = 64$, 64^3, 200 particles, `-near-shell fast` ($R = 3$, tile 4, order 5), 248 s on the A30.

7 Cross-validation and checkpoint transport

Two independent checks constrain the method externally: a cross-validation against PIConGPU's exact-quadrature field reference, and a checkpoint and restart transport. Neither is a truth comparison.

7.1 Cross-validation against the PIConGPU reference

The reference is the TSC-convolved field (Section 6), evaluated in PIConGPU [Burau et al., 2010] in its reference-only field-initializer mode (this work). `ctest` passes 10/10, and both sides use identical Yee

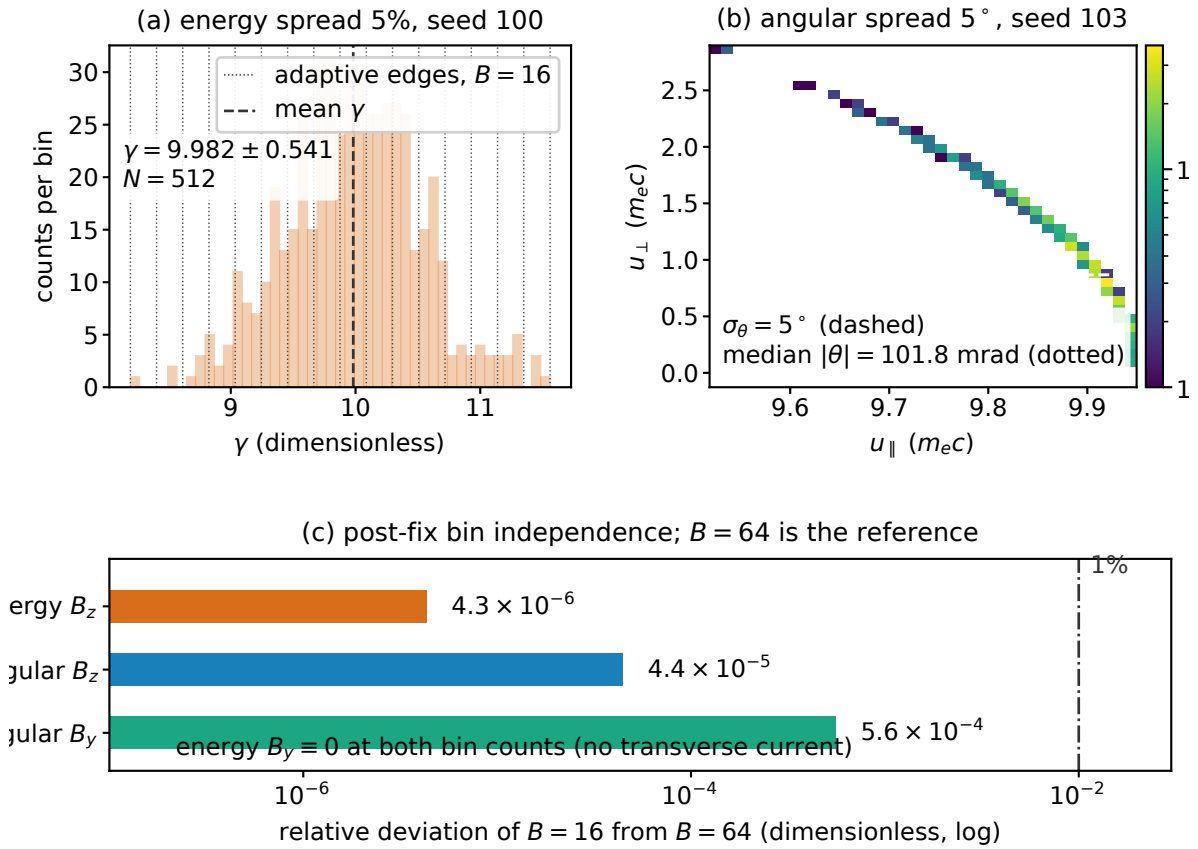


Figure 21: Sampled multi-velocity bunches ($\gamma = 10$, 512 sources, seeds 100 and 103); the $B = 16$ edges in (a) and the velocity boxes that produce the field come from the same `VelocityBinning` per-axis min/max path. (a) γ histogram for the 5% energy spread with the 16 adaptive edges drawn and the mean and standard deviation annotated; (b) $u_\parallel$–$u_\perp$ density for the 5° angular spread with the nominal 5° divergence and the measured median $|\theta| = 101.8$ mrad marked. (c) Relative deviation of the $B = 16$ peaks from the $B = 64$ reference, 4.3×10^{-6} (energy B_z), 4.4×10^{-5} (angular B_z) and 5.6×10^{-4} (angular B_y), with $B_y \equiv 0$ for the energy case, so the post-fix field is bin-independent at the stated $2.32050 \times 10^4 / 2.32049 \times 10^4$ T and 5.22×10^{-2} ratio.

offsets, grid, and origin; the reference staggered point maps to a picBunch node by index with zero snap deviation.

At 318 staggered points on a $64 \times 64 \times 32$ grid, cell 10^{-7} m, the overall worst per-component relative error is 4.86×10^{-2} for $\gamma = 1$ (E/z) and 6.20×10^{-1} for $\gamma = 10$ (E/z; E/y is 5.16×10^{-1}), against a reference reproducibility floor of 1.19×10^{-6} (E) and 1.43×10^{-6} (B) (this work). picBunch is about four to six orders of magnitude above that floor; the measured error is concentrated at the core and the beamed peaks, where the TSC support is, while the bulk agrees to below 1–3% and B is exactly zero for $\gamma = 1$ in both

The whole-field comparison, $N = 1000$ on $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes per component, gives the same picture: $\gamma = 1$ bulk median 3.8–7.8 $\times 10^{-3}$ with an L2 of 4.6–5.7 $\times 10^{-2}$; $\gamma = 10$ bulk median 1.4–3.3 $\times 10^{-2}$ but a beamed-core overshoot that pushes the E/y, E/z L2 to 0.76–0.92 (this work). The forward E/x is well resolved at $\gamma = 10$ ($\sim 6\%$); the beamed transverse E/y, E/z are 50–130% worst at the core, where the Lorentz-contracted field is a sub-cell pancake below the TSC support and the spectral near-field kernel resolution

Caveat: the sampling groups are distance-from-centroid. The 318-point groups (inside-support, face, outside) were cut by distance from the bunch centroid, in a bunch with a $\sim$ 6-cell spread (this work). That grouping may understate how close the "outside" points actually are to a source's TSC support, so the per-group numbers are a spatial profile, not a clean inside/outside-of-support split. The per-source dependence in Section 6 uses the explicit support union and is the measurement that separates the

Table 17: Whole-field comparison of picBunch against the PIConGPU `referenceOnly` field, $N = 1000$, $32 \times 32 \times 16$, cell 2×10^{-7} m, 16384 nodes/component, rosi_v100 (this work). `worst_rel` is peak-normalized; median and p99 are per-point over nodes with $|\text{ref}| > 10^{-3} \times$peak; L2 is pooled. Not a truth comparison.

component	worst_rel	median	p99	L2
$\gamma = 1$ (B identically 0)				
E/x	1.48×10^{-1}	3.92×10^{-3}	3.20×10^{-1}	4.62×10^{-2}
E/y	1.49×10^{-1}	3.76×10^{-3}	2.93×10^{-1}	4.75×10^{-2}
E/z	1.22×10^{-1}	7.81×10^{-3}	3.65×10^{-1}	5.67×10^{-2}
$\gamma = 10$ (B/x identically 0)				
E/x	1.36×10^{-1}	4.80×10^{-3}	1.23×10^{-1}	4.44×10^{-2}
E/y	1.98	1.36×10^{-2}	7.25	7.63×10^{-1}
E/z	1.93	3.33×10^{-2}	10.1	9.19×10^{-1}
B/y	4.42×10^{-1}	3.12×10^{-2}	1.41	2.58×10^{-1}
B/z	4.63×10^{-1}	1.52×10^{-2}	0.93	2.20×10^{-1}

per-source floor from any superposition contribution

7.2 Checkpoint format and bitwise arrival

The checkpoint writer emits a PIConGPU openPMD field checkpoint matching PIConGPU's own plugin schema: root layout, openPMD version 5.0 [Huebl et al., 2015], E, B $x/y/z$ at the exact Yee positions, grid spacing and offset, and the `picongpu_idProvider` mesh metadata (this work). The default precision is uniform `float32` for every field and particle record, so a stock single-precision PIConGPU can load the file `v0.7.0` copies every species from the input bunch and carries a single checkpoint-wide `picongpu_idProvider` range (this work); the `charge` and `mass` records are written rather than omitted, and the species is a single particle patch (Section 3.1). The remaining differences against a genuine PIConGPU checkpoint are non-blocking metadata (this work). For a multi-rank restart the writer splits the species into one patch per output rank: `patch_count` in `[placement]` gives cumulative, disjoint `numParticlesOffset` ranges, with one-cell overlap between neighbouring z regions, and every particle belongs to exactly one patch

Field arrival is bitwise. A fresh `v0.7.0` checkpoint written from a two-species input (16 electrons and 16 ions, `picongpu_idProvider` startId 0, nextId 32) restarts in PIConGPU with exit 0 under both `patch` modes (`bbox`: offset $(7, 8, 3)$, extent $(8, 10, 6)$; `domain`: offset $(0, 0, 0)$, extent $(24, 24, 12)$), all six field components bitwise-equal, $\max_{\text{abs diff}} = 0.0$, electrons with ids 0 to 15, first-step displacement exactly $\beta c\,dt$ with $\beta = 0.9949874367$ (this work). A validation-only build with the field solver set to `None` loads the checkpoint and writes step 1; all six components are bitwise-equal, max ULP 0

A multi-rank restart with the four-patch layout was verified process-level: a 256-particle bunch written as four non-empty patches (cumulative offsets $0, 56, 126, 203$, one-cell overlap) restarts PIConGPU with exit 0, and after one step all six field components are bitwise-equal to the checkpoint, $\max_{\text{abs diff}} = 0.0$

7.3 PML passthrough and open-boundary restart

On a $24 \times 24 \times 12$ deck the restart is exit 0 with `-periodic 1 1 1`, and the 16 electrons load; initialization drops from the ~ 116 s neuralBunch path to 1.8 s because the checkpoint carries the field (this work). For an open-boundary deck the writer copies the `Convolutional PML` slabs from the input checkpoint verbatim (converting a non-float32 slab to float32), so the PML state arrives with the restart. On a real LWFA checkpoint (638976 electrons, eight `Convolutional PML` groups with six datasets each, $64 \times 128 \times 64$, `-periodic 0 0 1`) the writer copies all eight groups and the restart exits 0; the copied PML datasets are float32 and bitwise equal to the source, initialization is 3.986 s, and the main E and B records are recomputed from the particles and stay float32

8 Discussion

The method computes the present-time field of a uniformly moving, TSC-shaped bunch as a free-space FFT convolution. It is a far-field method.

What it is good for. In the far field the relative error against the point field is below 10^{-3} on the beam axis from $d = 6$ for every γ. Against the TSC-convolved continuum: $\gamma = 1$, aspect 1 crosses 3×10^{-4} at $d = 12$ (1.1×10^{-4}); $\gamma = 1$, aspect 4 at $d = 32$ (2.6×10^{-4}); $\gamma = 10$, aspect 1, is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$ At 2×10^6 particles in a 20-cell box the compute-only total is 64.8 ms, dominated by the deposit (56.5 ms), and flat in γ Repeated runs are bit-identical (md5 `d3641d64...`); the Coulomb limit reproduces the TSC-convolved Coulomb field to 4.88×10^{-14}.

Measured error in the bunch core. The mesh alone, on the $N = 1000$ slice geometry, has a worst per-component plane error of 0.12–0.415 for $\gamma = 1$ and 0.11–4.60 for $\gamma = 10$ where the planes cut the bunch core (Table 11); those core errors are measured against the order-4 driver, which agrees with the order-12 arbiter to within a few percent at the worst in-support nodes of that $32 \times 32 \times 16$ geometry, so the driver is adequate there (Section 6.5). The exact near shell (Section 3.7) is opt-in and its assembly is exact to roundoff; at the measured $R = 3/4$ radii on the $h = 1$ slice grid the residual is the mesh contribution outside the shell, not the tile subtraction (Section 6.7). The arbiter-validated in-support accuracy of the fast backend is 1.6% at the closest moving node, not the 0.31 the driver-based slices reported (Section 6.5, this work). In the perpendicular direction at $\gamma = 10$, aspect 1, the measured $M-T$ error at $d = 48$ is 2.6×10^{-4}, the $M-P$ error is 1.5×10^{-2}, and the TSC smoothing $T-P$ is 1.48×10^{-2}; the $M-P$ error is the TSC smoothing, not the deposit (Table 8). At $r = 8$ cells the node spacing for 10^{-3} perpendicular error at $\gamma = 10$ is $h \approx 1/29$, a 2.4×10^4-fold increase in M^3 FFT work relative to the $h = 1$ run. At aspect 4, $\gamma = 1$, the perpendicular $M-T$ error at $d = 48$ is 2.4×10^{-3}. The cross-validation confirms the same structure: the measured disagreement with the exact reference is per-source, local to the TSC support, and independent of particle count.

Performance. The cost is two regimes with a crossover set by the box, not the particle count. At large N the deposit dominates, linear in N, 52–57 ms on the 2×10^6 rows. At large boxes the FFT and kernel dominate: from $M = 60$ to $M = 220$ the total goes from 64.8 ms to 233 ms, and halving h costs 1.07× and 2.35× The compute timings are single-GPU on one `rosi_v100` host; the same mesh is 0.052 s on one H100 (Section 4) The rank scaling is measured on an idle V100 window: the component-parallel mesh plateaus at 1.20× (replicated deposit and source FFT), the tile-parallel near shell at 1.34× (static tile split plus all-reduce), and the collective writer improves to 1.71× at two ranks and falls to 0.48× at four, where it is I/O-bound (Section 4.1). The near-field cost is a separate wall, measured end-to-end through the CLI on one H100: 281.0 s for the support-only shell ($R = 1.5$) and 1510.7 s for the four-cell shell ($R = 4$) at 2×10^6, against a mesh-only wall of 2.48 s (Section 6.8). The shell radius is the dominant cost lever, and the device backend runs the same fast algebra on the CUDA device, agreeing with the host fast path to 5.4×10^{-16}–8.0×10^{-16} (Section 6.9); the distributed FFT makes the output field slab-local at the cost of `MPI_Alltoallv` transposes that are 74–85% of the FFT-plus-transpose time, a memory feature rather than a speed feature (Section 6.10).

Measured error by configu-ration. In floored relative per-component error against the TSC-convolved continuum, the measured values at $d = 12$ are 1.1×10^{-4} for $\gamma = 1$ aspect 1 and 2.6×10^{-4} for $\gamma = 1$ aspect 4; the axial component is below 3×10^{-4} from $d = 6$ for every γ. The perpendicular direction at $\gamma = 10$, aspect 1, is 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$, the measured values at those distances The fast near backend, against the same continuum, is 7.99×10^{-4} stationary interior, 1.96×10^{-5} face, 1.57×10^{-7} outside; 1.69×10^{-4} moving axial interior; and 8.15×10^{-2} moving oblique interior, the measured limit of the fast backend (Section 6.6). Each statement in Section 6 names which metric (floored, unfloored, absolute) supports it; the floored column is electric-only at $\gamma = 10$ because the floor excludes the magnetic field, and the judge metric $|\text{ref} - \text{mesh}|/(1 + \max)$ reports $d = 2$ for every case and does not discriminate. The radii in Table 10 are axis directions only; oblique directions are $\sim 1.5\times$ larger. At a fixed distance in cells the controlling length is the beaming cone width d/γ; for $\gamma \geq 50$ the cone is sub-cell and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls to about 1 (Section 6.13).

Structural assumptions. The method assumes a single shared velocity (or a set of binned velocity classes, one convolution per class, Section 6.16); the near shell needs no binning because each pair carries its own velocity. The computed quantity is the TSC-convolved uniformly-moving point-charge field; the convolution is free-space (no boundary condition other than zero fill beyond the far-tail switch); and there is no radiation term, because the motion is uniform. A bunch with a velocity spread requires one

convolution per velocity class; the measured binned behaviour (v0.9.1, bin-independent to 0.004% for the 5% energy spread and 1.4% against the arbiter for the 1 mrad case) is in Section 6.16, and the validated shipped scope is a single shared **u** (Section 2).

9 Conclusion

`picBunch v0.7.0` computes the present-time field of a uniformly moving, TSC-shaped bunch as a zero-padded free-space FFT convolution with staggered Yee read-out, driven by a single TOML configuration (Section 3.1), and writes a uniform float32 checkpoint that restarts under a stock PIConGPU, periodic and open-boundary, single- and multi-patch. The release passes 18/18 ctest with zero compiler warnings on a V100 `sm_70`, an H100 `sm_90`, and a gfx1100 HIP build

The opt-in near shell assembles exactly to roundoff, and its tile subtraction is not the limiting term; the seam excess at the fast/exact crossing is 2.03×10^{-10} per component for a stationary single particle (2.8×10^{-10} for the 64-source cloud). The fast backend, an eight-term `asinh` box plus a plain Gauss–Legendre remainder, meets 10^{-3} stationary (7.99×10^{-4} interior) and moving axial (1.69×10^{-4} interior); the moving oblique interior is 8.15×10^{-2}, the measured limit of the fast backend At 2×10^6 particles the mesh-only CLI wall is 2.48 s on one H100 (compute 0.116 s), deterministic (md5 `d3641d64...`); the near shell is 281.0 s end-to-end support-only at 7.99×10^{-4} and 1510.7 s end-to-end four-cell at the same interior error, the shell radius the dominant cost lever (Section 6.8). The device backend runs the fast algebra on the CUDA device, agreeing with the host fast path to 5.4×10^{-16}–8.0×10^{-16} (Section 6.9), and the distributed FFT makes the output field slab-local with the transposes dominating the FFT-plus-transpose time, a memory feature (Section 6.10).

In the far field the mesh is below 3×10^{-4} from $d = 12$ cells for $\gamma = 1$, aspect 1, and from $d = 32$ for $\gamma = 1$, aspect 4; at $\gamma = 10$, aspect 1, the measured values are 6.5×10^{-3} at $d = 32$ and 1.8×10^{-3} at $d = 48$. The γ-landscape measurement shows the error growing steeply with γ at a fixed distance in cells, with the beaming cone width d/γ the controlling scale: at $d = 8$ cells, $h = 1$, the error is 4.4×10^{-2}, 1.05, 5.47 for $\gamma = 10, 50, 100$, and the h-refinement exponent p (error $\sim h^p$, fitted from the two spacings) falls from 1.85 to about 1.0 as γ rises (Section 6.13). The MPI compute is component-parallel mesh and tile-parallel near shell with one all-reduce each, bitwise identical at 1/2/4 ranks, with the replicated deposit and source FFT capping the mesh speedup and the collective writer I/O-bound at four ranks (Section 4.1). The method is a far-field tool: cheap, deterministic (md5 `d3641d64...`), and portable across the three backends; the measured in-support error at $\gamma \geq 50$ is reported in Section 6.13.

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